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Cutting the energy bills of Internet Service Providers and telecoms through power management: An impact analysis

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ABSTRACT

The energy consumption of the Information and Communication Technology (ICT) sector has been increasing recently; this sector is estimated to account for 2% of the total energy consumption. An even more aggressively increasing trend is the volume of Internet traffic and the number of connected devices. Thus, reducing the energy needs of the Internet is recognised as one of the main challenges that the ICT sector will have to face in the near future to reduce its overall energy footprint. Introducing energy-efficient techniques, both at the device level and the network level, is required.

The main goal of this work is to quantitatively evaluate the potential energy savings from applying energy-efficient techniques, while examining the trade-off between network performance and the achieved energy savings.

We introduce a categorisation of the energy-aware design space, focusing on the existing techniques in the device data plane, and contribute an analytical framework to represent the impact of energy-aware technologies and solutions for network devices. Our energy profile model represents the diverse energy-aware states of the network devices and is applied over two reference scenarios, one of a large-scale Telco (Telecom Italia) and one of a medium size Internet Service Provider (GRNET), to evaluate the impact of each energy-aware technology and the energy savings potential at the Home, Access, Metro/Transport and Core parts of each network.

The results show the estimates of energy savings exceed 60% in many cases, while maintaining the same quality of service as in the energy-agnostic case.

1. Introduction

The energy consumption of the Information and Communication Technologies (ICT) sector has followed an increasing trend in the last several years and it is estimated to have accounted for 2% of the total energy consumption in 2007 [1]. Because of the way energy is still being produced currently, most energy consumption is accompanied by non-negligible Green House Gas (GHG) emissions that have severe consequences on climate change. In a Business-As-Usual (BAU) scenario, the GHG emissions of the ICT sector are expected to reach 1.43 Gtons carbon dioxide equivalent (CO_2e) in 2020 (they amounted to 0.53 Gtons in 2002) [2]. On the other hand, taking global predictions into account, a decrease of between 15% and 30% in the emitted volume would be required before

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2020 to keep the global temperature increase at less than 2 °C [3].

The increasing trend in the energy consumption of the ICT sector has also been confirmed in recent reports published by large Telecom operators (Telcos) and Internet Service Providers (ISPs) worldwide. In 2006, the overall energy consumption of Telecom Italia had already reached more than 2 TWh (approximately 1% of the total Italian energy demand), which was an increase of 7.95% over 2005 [4–6]; in 2009, this consumption increased to 2.14 TWh. Another representative example is from British Telecom, whose overall power consumption for its network and estate during the 2010 financial year was 3.12 TWh [7,8]. The Deutsche Telecom group reported an overall energy consumption of 7.91 TWh during 2009, compared with 3 TWh in 2007 [9], and this group attributes the steep increase to technology developments, increasing transmission volumes and network expansion. The power consumption of Verizon in 2010 was 10.24 TWh, up from 8.9 TWh (approximately 0.26% of USA's energy requirements) in 2006. AT&T accounted for 11.14 TWh in 2010 and declared a consumption of 654 kWh per terabyte of data carried on its network in 2008. The requirements of France Telecom were approximately 4.38 TWh in 2009 [10], while it is committed to a 15% reduction in its global energy consumption by 2020 relative to the 2006 level. In 2010, the Telefonica group consumed 6.37 TWh, compared with their 2006 figure of 1.42 TWh, which amounts to 0.6% of Spain's total energy consumption [11]. The NTT group reported that the amount of electrical power needed for telecommunications in Japan was 4.2 TWh in fiscal year 2004 [12] and that their direct energy consumption amounted to 2.75 TWh in 2009. Finally, China Mobile's energy requirements since 2009 have exceeded 10 TWh, an annual increase of more than 1 TWh. These and other data on the yearly energy consumption of some of the major telecom operators world-wide are presented in Table 1.

In parallel with the increase in energy consumption, an even more aggressive trend is observed in Internet traffic, as well as in the number of devices connected to the Internet. Worldwide, the growth rate of Internet users is approximately 20% per year, while in developing countries, this growth rate is close to 40–50% [13]. According to the Cisco Visual Networking Index [14], global IP traffic has increased eightfold over the past 5 years and it will increase fourfold over the next 5 years. Overall, IP traffic is estimated to grow at a compound annual growth rate of 32% from 2010 to 2015, with busy-hour traffic growing more rapidly than the average rate. It is important to note that in 2010, global Internet video traffic surpassed global peer-to-peer (P2P) volume, and by 2012 Internet video is predicted to account for over 50% of consumer Internet traffic [14].

Considering these facts, as well as predictions for the near future, a reduction in the total energy consumption and, consequently, in the carbon footprint of the ICT sector appears to be quite desirable. According to GeSI [2], if applied to improve the energy efficiency in other sectors, ICT is seen to have the potential to eliminate approximately 15% of the global carbon footprint. *A fortiori*, ICT should apply the same energy efficiency awareness to its own operations; this is indeed happening, due also to the economic constraints that stem from the need to counterbalance the ever-increasing cost of energy with the desired level of performance of network and computing devices. However, to achieve this kind of reduction in the energy consumption of next generation networks and devices, new paradigms have to be introduced in both design and daily operation phases. As a relevant example for this paper, the energy efficiency of network devices needs to be significantly improved and, in parallel, advanced mechanisms have to be developed for optimal network operations relative to the total energy consumption. Thus, there is a need to introduce energy-efficient techniques both at the device level and at the network level.

Such techniques need to be developed and applied because the design of the Internet and even recent evolutions do not address energy consumption issues [15]. Today, overprovisioning is adopted in dimensioning both the number and the capacity of network devices and links. The degree of overprovisioning depends on the maximum load that the network has to carry during its peak hours, which is directly connected with the peak energy consumption of the network, as well as with the redundancy degree of the network. This degree of overprovisioning is selected for increased resiliency and fault-tolerance of the current networks. In addition to the redundancy level of devices and links in the network that increase the total energy consumption, any drop in traffic is not followed by a corresponding drop in the amount of energy consumed by the network, which is largely due to the lack of energy proportionality in the current generation of network devices, which consume close to their peak energy independently of their actual traffic load. Improving hardware (HW) efficiency only would not be enough to reduce the impact of energy consumption; indeed, even though data from network device manufacturers show capacities continually increasing by a factor of 2.5 every 18 months, the energy efficiency of silicon technologies improves at a slower pace, in accordance with Dennard's law (i.e., by a factor of 1.65 every 18 months) [16].

Starting from the above considerations, it should be clear that reducing the energy consumption of the Internet

Table 1
Yearly energy consumption of some of the major telecom operators worldwide. These data were obtained by the sustainability report of each company.

	Energy consumption (TWh per year)				
Telecom	2006	2007	2008	2009	2010
Deutsche Telekom (World)	7.10	7.22	7.84	7.91	–
France Telecom (World)	3.66	3.47	4.57	4.38	–
Telecom Italia	2.10	2.15	2.13	2.14	–
British Telecom (UK)	1.94	1.99	2.03	2.28	2.28
British Telecom (World)	–	–	2.6	2.71	3.12
AT&T (World)	–	–	–	11.07	11.14
Verizon	8.90	–	–	10.27	10.24
NTT	–	2.76	2.76	2.75	–
Telefonica	1.42	–	4.76	5.05	6.37
SwissCom	–	–	0.43	0.40	0.40
China Mobile	–	–	9.35	10.62	11.94
SK Telecom	–	–	0.94	1.09	1.09

might be one of the main challenges that the ICT sector will have to face in the future. Thus, the main objective of the Energy COnsumption NETworks (ECONET) project [5] is to design and develop innovative solutions and device prototypes for wired network infrastructures (from customer premises equipment to backbone switches and routers) by 2013. The resulting network platforms will adopt green networking technologies for aggressively modulating power consumption according to the actual workloads and service requirements. Within the framework of the ECONET project, the work presented in this paper aims at evaluating the impact of specific ECONET targets in single devices and their subcomponents on the energy consumption of overall networking infrastructures under two realistic reference scenarios. The main goal is to show that the ECONET technical objectives will be sufficient to achieve approximately 50% energy savings in the short-term and 80% in the long-term. This paper introduces an analytical framework to represent the impact of energy adaptive technologies and the solutions for network devices. This modelling framework is an extended version of the one proposed in [17] and aims to predict the energy gains achievable using both standby and dynamic adaptation primitives. The impact analysis proposed here has to be considered complementary to the ones proposed by Tucker et al. [18–21] because we do not focus on the energy consumption of "inertial" improvements in networking technologies, but we try to understand the potential impact of power management primitives, which are almost totally absent in today's network devices and protocols.

The paper is organised as follows; Section 2 categorises the energy-aware design space, focusing mainly on the existing techniques in the device data plane. Section 3 details the energy profile model for representing the diverse energy-aware states of network devices, while Section 4 briefly presents the two reference scenarios, emphasising the energy consumption and the representative traffic profiles of the considered networks. In Section 5, a detailed evaluation of the energy-aware mechanisms is provided based on the application of the designed model in the reference scenarios. Finally, Section 6 concludes the paper with a short summary of our work and a discussion of the open issues and future directions.

2. Energy-aware design space

Improving a network's energy efficiency requires the synergy of several technologies using energy consumption as the primary objective. A comprehensive treatment of this multidimensional problem is necessary to provide a solution that offers considerable savings [22].

The energy-aware design space is divided into energy-aware technologies in the data and the control plane. The most crucial energy-related improvements in network equipment design mainly refer to the data-plane of network equipment because this plane usually includes the most energy starving hardware elements. Energy-aware technologies in the data plane refer to the design of novel network-specific capabilities that are able to optimise the power management features (e.g., standby and power scaling primitives) in device architectures, by meeting in parallel their network operational constraints. On a second level, exploiting the equipment's novel green capabilities/functions forms another important design area, when considered together with the overprovisioned design of network infrastructures; this level of consideration allows the development of local and network-wide control strategies to minimise the overall energy consumption while meeting the operational needs and can be regarded as the design of energy-aware strategies at the control plane. Thus, energy-aware technologies in the data plane can constitute the basis for the design of broader network-wide control strategies for the energy efficient operation of a network as a whole.

In this section, these two areas of energy-aware design recognised as the most crucial – energy-aware design of the network device data plane functions and energy-aware strategies' design at the control plane – are further explained. It should be noted that, in this paper, our focus is mainly on the energy-aware technologies in the data plane.

2.1. Energy-aware design of the network device data plane functions

According to [22], at the highest level, energy-related improvements in network equipment design can be classified as organic and engineered. Organic efficiency improvements are commensurate with Dennard's scaling law; every new generation of network silicon packs more performance in a smaller energy budget. Engineered improvements refer to active energy management including, but not limited to, idle state logic, gate count optimisation, memory access algorithms, I/O buffer reduction and so forth.

In this framework and as outlined in [16], the largest part of current approaches for engineered improvements is founded on a few base concepts, which have been generally inspired by energy-saving mechanisms and power management criteria that are already partially available in computing systems. These base concepts for energy efficiency in wire-line networks can be classified into Re-engineering, Dynamic Power Scaling (DPS) and Sleeping/standby approaches.

Re-engineering approaches aim to introduce and design more energy-efficient elements for network device architectures, to suitably dimension and optimise the internal organisation of devices, as well as to reduce their intrinsic complexity levels [23].

The dynamic power scaling of network/device resources is designed to modulate the capacities of packet processing engines and network interfaces to meet the actual traffic loads and requirements. Power scaling can be performed using two power-aware capabilities, namely, dynamic voltage scaling and idle logic, which both allow the dynamic trade-off between packet service performance and power consumption.

Finally, sleeping/standby approaches are used to smartly and selectively drive unused network/device portions to low standby modes and to wake them up only if necessary. However, because today's networks and related services and applications are designed to be continuously

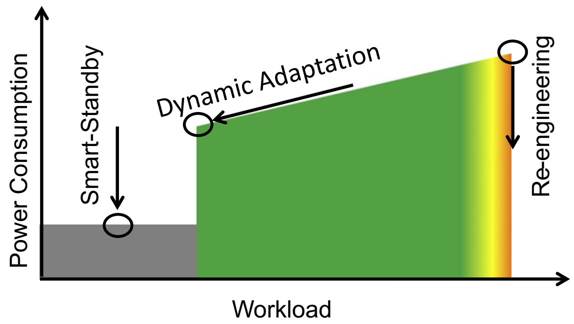


Fig. 1. The power profile of future Internet devices and the roles of re-engineering, dynamic adaptation and smart standby technologies.

and always available, standby modes have to be explicitly supported with special proxying/virtualisation techniques that are able to maintain the "network presence" of sleeping nodes/components and to guarantee a short "wake-up" time. These approaches are not mutually exclusive and research efforts will be needed in all of these areas to effectively develop next-generation green devices.

In more detail and as shown in Fig. 1, these approaches will together allow defining an "energy profile" for next-generation network devices. Re-engineering will result in reducing the maximum power consumption of devices and dynamic adaptation will enable the modulation of energy absorption according to the actual workload. Finally, techniques based on smart standby will help to further cut the power consumption of unused devices (or parts of them) by allowing HW components to enter very low power sleeping states.

The adoption of DPS and smart standby optimisation obviously affects network performance, given that these approaches are founded on the ideas of tuning device processing/transmission capacities and of waking up the hardware upon "work" request. In other words, the use of such green optimisation methods allows the trading of energy consumption for network performance (often measured in terms of packet processing/transmission delay).

2.1.1. Dynamic power scaling

Power scaling capabilities allow the network to dynamically reduce the working rate of processing engines or link interfaces by adopting two basic techniques, namely, *Adaptive Rate* (AR) and *Low Power Idle (LPI)*. The former allows the dynamic modulation of the capacity of a link, or of a processing engine, to meet traffic loads and service requirements. The latter forces links or processing engines to enter low power states when not sending/processing packets and to quickly switch to a high power state when sending one or more packets. These techniques are not exclusive and can be jointly adopted to adapt the system's performance to the current workload requirements.

LPI and AR have different impacts on packet forwarding performance. As shown in Fig. 2, AR (Fig. 2c) obviously causes a stretching of packet service times, while the sole adoption of LPI (Fig. 2b) introduces an additional delay in packet service due to the wake-up times. Moreover, preliminary studies in this field [24–26] showed how performance scaling and idle logic work like traffic shaping mechanisms by causing opposite effects on the traffic burstiness level. The wake-up times in LPI favour packet grouping and then an increase in traffic burstiness, whereas service time expansion in AR favours burst untying and, consequently, traffic profile smoothing. Finally, as outlined in Fig. 2d, the joint adoption of both energy-aware capabilities may not lead to outstanding energy gains because performance scaling causes larger packet service times and, consequently, shorter idle periods. It is worth noting that the overall energy saving and the network performance strictly depend on the incoming traffic volumes and statistical features (interarrival times, burstiness levels, etc.). For instance, idle logic provides top energy- and network-performance when incoming traffic has a high burstiness level because fewer active-idle transitions (and wake-up times) are needed and the HW can remain in its low consumption state for longer periods.

In previous work [27–29], Gupta et al. faced the energy efficiency issue in wireline networks and developed several algorithms that used traffic prediction to put links to low power idle modes. The idea behind these algorithms is simple – the upstream interface on a link maintains a window of packet interarrival times. This information is then used to determine the length of time that an interface can be put to sleep, such that, with a high probability, the buffers at that interface will not overflow. The constraint is the non-zero time it takes for the downstream interface to wake up. The results obtained with real traces show that,

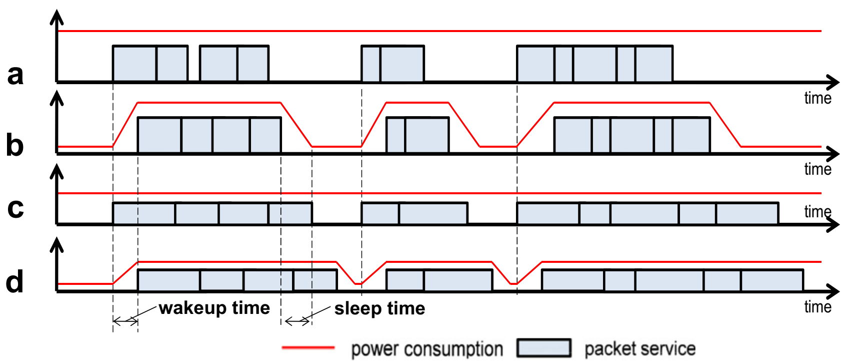


Fig. 2. Packet service times and power consumptions in the cases with: (a) no power-aware optimisations, (b) only LPI, (c) only AR, (d) AR and LPI.

for loads up to 30% of link capacity, considerable energy savings can be achieved.

Similarly, research by Christensen et al. has specifically addressed how to reduce the direct energy use of Ethernet links and has contributed to the development of the IEEE 802.3az standard [30]. In [31], they first explored the notion of Adaptive Link Rate (ALR) for the Ethernet, whereby a link would operate at a low data rate during periods of low utilisation and at high data rate only for high utilisation periods. Given that most links are highly underutilised, with ALR, most Ethernet links could operate at a low data rate (and, thus, reduce their energy consumption compared to operating at a high data rate) most of the time [32].

An implementation of ALR would entail an Ethernet interface having two physical layers and switching between them. The time to switch between physical layer implementations was deemed to be a major issue, resulting in an alternative LPI approach [33] proposed by Intel. LPI is the approach specified in the IEEE 802.3az standard and currently allows a 10 Gbps link to wake up in less than 3 μs.

In [34], Bolla et al. analysed and empirically modelled the energy modulation capabilities of processing engines in Linux-based software routers equipped with general-purpose and multi-core processors that already included LPI and adaptive rate primitives. The results were obtained by evaluating several hardware architectures and they suggest that such technologies permit the near-linear scaling of the trade-off between power consumption and network performance.

In [35], the authors extended their approach by introducing a control framework for optimally tuning LPI and adaptive rate mechanisms to statistically meet the current traffic loads and service requirements. The results obtained on real traffic traces show that energy gains up to 60% are feasible.

Working on similar platforms, Intel researchers [24] specifically focused on LPI primitives and performed a comprehensive study of the impact of transition times on LPI as a function of the load. They showed that, as the transition times shrink from a value of 10 ms to 1 ms and then further to 100 μs, the time spent sleeping at 30% load goes from 0 at transition time of 10 ms to 40% when this time is 1 ms and to 70% when the transition happens in 100 μs.

2.1.2. Smart standby

Sleeping and standby approaches are founded on power management primitives that allow devices or parts of them to turn themselves almost completely off and enter very low energy states, while all their functionalities are frozen. The widespread adoption of such energy-aware capabilities is generally hindered by the necessity to maintain the "network presence" of the device. Network hosts and devices must maintain network connectivity, or else they would literally "fall off the network" and become unreachable, causing network applications and services to fail.

In [36,37], Christensen et al. first explored the idea of using a proxy to "cover" for a network host to allow it to go to sleep. Specifically, they addressed the requirements for a proxy to be able to respond to Address Resolution Protocol (ARP) packets on behalf of a sleeping host (to maintain reachability from the router) and to respond to other protocol and application messages as needed to maintain full network presence. A specific focus of their research was on P2P protocols and how they might be proxied to allow for a PC sharing files to sleep most of the time. The proxying requirements they developed led to the creation of the ECMA TC38-TG4 standard. Apple, Inc. has recently announced a product with green proxying capabilities.

The development of such network-specific low-energy modes is a fundamental key factor for reducing the carbon-footprint inside networks because it will allow switching some links, entire network devices, or parts thereof to sleep mode in a smart and effective way.

2.2. Energy-aware strategies design at the control plane

Designing control strategies from the energy perspective is the final step towards improving a network's energy efficiency because the network and the power-aware capabilities of the network components are taken into account to devise energy-saving mechanisms. Currently, network design is based on deploying and maintaining densely interconnected infrastructures built for reliability and high performance. Typically, high-end routers are used in the core, while lower-end distribution routers are placed around the core and even lower-end access routers and switches are used at the periphery. Understanding the power demands of current networking equipment under different configurations and traffic loads provides the opportunity to replace power-hungry systems with lower power systems that still provide the required reliability and performance.

Virtualisation technology is a powerful tool to help with this goal. Virtualisation refers to the division of a hardware platform into several isolated virtual environments. The virtualisation capabilities of network equipment [38], such as routers and switches, are the basis for developing network consolidation strategies to achieve a more efficient utilisation of resources while permitting the network to operate fewer energy-consuming physical devices.

Another area of energy-aware control strategies design is that of energy-aware traffic engineering approaches [39]. These approaches, as an extension of traditional traffic engineering with a new primary objective, are targeted to effectively serve traffic, while reducing the network's overall energy footprint. They are aimed at dynamically reducing parts of the network to low power or sleep states during light utilisation periods to minimise the energy requirements of the overall network, while meeting the operational constraints and current workloads [40–46]. Information about the network characteristics, traffic profiles analysis, as well as the capabilities of the equipment to change operating state, is exploited to minimise the power usage.

Finally, resource allocation strategies [47] and mechanisms for improving the algorithmic efficiency of the developed protocols [48] are part of the energy-aware design space in the control plane because they directly affect energy consumption and provide the potential for further energy saving.

3. The energy profile model

This section introduces the analytical framework for modelling the impact of energy adaptive technologies and solutions for network devices under investigation by the ECONET consortium. This modelling framework is an extended version of the one proposed in [17] and aims to predict the energy gains achievable by using both standby and dynamic power scaling primitives. In the case of DPS primitives, both idle logic and adaptive rate are explicitly considered.

The entire model is specifically designed to capture the main effects and potential benefits of the energy saving primitives in a highly parametric way without entering into the details on how these primitives will be realised inside device platforms. We modelled each energy-aware primitive using an extended set of efficiency and shape parameters, which allows us to go beyond the simplistic representation of the device energy profile shown in Fig. 1 and used in [17].

We believe that these parameters capture the main peculiarities of the energy-aware primitives under investigation. Furthermore, in this paper (Section 5), this extended parameter set will allow us to understand how and which green primitives have to be designed to maximise their impact on the existing network infrastructures.

Recalling the simple representation in Fig. 1, we can note that the energy profile of a network device can be roughly described using four main parameters:

- the maximum energy consumption $\Phi_{a\max}$ when the device is fully active and operating at the maximum speed (in this case, $\Phi_{a\max} = \Phi$, where Φ corresponds to the energy consumption of the device with no green technologies enabled);
- the minimum energy consumption $\Phi_{a\min}$ when the device is active, but operating at the minimum speed;
- the energy consumption $\Phi_{i\min}$ when the network device is sleeping in the idle state;
- the energy consumption Φ_s when the device (or parts of it) is sleeping in standby mode.

As already proposed by the ECONET consortium and in [17], we consider that $\Phi_{a\max} \geqslant \Phi_{a\min} \geqslant \Phi_{i\min} \geqslant \Phi_s$; more specifically, because these optimisation mechanisms will be applied at data plane HW only:

$$\Phi_{a\max} = \Phi \tag{1}$$

$$\Phi_{a\min} = \Phi[\psi_{ctr} + (1-\beta)\psi_{data} + (1-\kappa)\psi_{cool}] \tag{2}$$

$$\Phi_s = \Phi[\psi_{ctr} + (1-\alpha)(\psi_{data} + \psi_{cool})] \tag{3}$$

The ψ_* parameters represent the device's internal breakdown of energy consumption sources. Parameter ψ_{ctr} represents the share of energy consumption due to the control plane hardware, parameter ψ_{data} represents the share of energy consumption due to the data plane hardware and parameter ψ_{cool} represents the share of energy consumption due to the cooling system and the power supply. Parameters α and β represent the efficiency degrees of the green technologies under investigation, e.g., the standby capability and the power scaling capability, respectively, while κ represents the indirect gain on the cooling system and the power supply. In this study, we consider that $\Phi_{i\min} = \Phi_{a\min}$ to provide a conservative estimate of energy savings. In Table 2, a detailed description of our notation appears.

The way with which the energy consumption changes between the $\Phi_{i\min}$ and Φ parameters according to the traffic load fluctuations will obviously depend on the typology, the parameterisation and the efficiency of power scaling primitives (i.e., AR and LPI). In this respect, the linear relationship between the workload and the energy consumption in Fig. 1 should certainly be considered as an ideal

Table 2
Notation used in the impact model.

Φ	Business-As-Usual (BAU) energy consumption of the device (i.e., without any energy-aware primitives enabled)
$\Phi_{i\min}$	Power consumption when the network device is sleeping in the idle state
Φ_{amin}	Minimum power consumption when the network device is active
Φ_{amax}	Maximum power consumption when the network device is active
Φ_s	Power consumption when the whole device or parts of it fully sleep in standby mode
ψ_{ctr}	Share of energy consumption due to control plane hardware
ψ_{data}	Share of energy consumption due to data plane hardware
ψ_{cool}	Share of energy consumption due to cooling and power supply
α	Efficiency level of the standby capability
β	Efficiency level of the power scaling capability
κ	Indirect gain on cooling system and power supply
δ	Efficiency level of green traffic engineering and routing policies
S	Number of available power states
$\mu^{(s)}$	Normalized value of maximum service capacity when the device is working in the sth state. Note that $\mu^{(S-1)} = 1$
λ	Normalized value of traffic load incoming to the device
v	Shape parameter of power states' energy consumption
σ	Shape parameter of idle optimisation efficiency (ideal case $\sigma = 1$, real case $\sigma < 1$)
γ	Dependency parameter of idle optimisation from power states (if $\gamma = 0$, idle logic is completely unaware of power states, if $\gamma = 1$ there is no idle logic)
$\Phi_a^{(s)}$	Maximum power consumption when the network device is active and working in the sth state
$\Phi_i^{(s)}$	Power consumption when the network device is idle in the sth state
$\widetilde{\Phi}^{(s)}$	Average energy consumption of a device using LPI and AR primitives and working in the sth power state
ξ	Conservative level of exploitation of green technologies
ω	Increase degree of traffic volumes in devices remaining active after the use of the green traffic engineering
$\widetilde{\Phi}(\lambda)$	Average energy consumption of an active device in the presence of incoming load λ
$\widetilde{\Phi}_{real}$	Average energy consumption of an active device according to a traffic volume with probability density distribution p_{real}
$p_\lambda(\lambda)$	Probability density distribution of incoming traffic load
$\boldsymbol{\Gamma}$	1×4 vector containing the number of devices per network layer (home/access/metro/core)
$\boldsymbol{\Omega}$	1×4 vector containing the average power consumption per device per network layer (home/access/metro/core)
$\boldsymbol{\theta_d}$	1×4 vector containing the average redundancy degree of devices per network layer (home/access/metro/core)
$\boldsymbol{\theta_l}$	1×4 vector containing the average redundancy degree of links per network layer (home/access/metro/core)

case because AR and LPI may introduce different aspects that may affect system performance [26].

For instance, the intrinsic nature of AR primitives leads the device to work with a finite, discrete set of stable HW configurations (in terms of input voltage, clock frequency speed, etc.), generally referred to as the "state". Depending on the HW technology in use (ASIC, FPGA, etc.) and on the implementation of AR primitives (e.g., dynamic frequency scaling – DFS, dynamic voltage scaling – DVS, dynamic frequency and voltage scaling – DVFS), the relationship between energy consumption and performance with respect to AR states may exhibit different trends (e.g., linear, quadratic, etc.).

As far as LPI primitives are concerned, we have to take the energy overhead, which is needed by the HW to enter and to re-wake up from low power modes, into account. As shown in [24,34], the more this additional energy is not negligible, the more it makes the profile follow a curve with increasingly pronounced concavity with respect to that of Fig. 1.

Due to all of these considerations, Sections 3.1 and 3.2 introduce a general model for estimating the energy profile of a device with AR and/or LPI primitives. Section 3.3 is devoted to calculating the optimal AR and LPI configurations to meet the incoming traffic load and service requirements. Section 3.4 focuses on the modelling and the usage of the sleeping/standby primitive.

3.1. The AR model

We assume the presence of S power-states. The generic sth power state, with $s = 0,\ldots,S-1$, is represented by a pair composed of the maximum forwarding capacity $\mu^{(s)}$ and the maximum energy consumption $\Phi_a^{(s)}$.

The list of power states is thought to be ordered in s: $s = 0$ corresponds to the state with the lowest energy consumption and forwarding capacity; $s = S-1$ is the most energy-hungry state, which provides the highest network performance. In more detail, the $S-1$ state corresponds to the case of a device without any AR primitives enabled; therefore, $\mu^{(S-1)} = 1$ and $\Phi_a^{(S-1)} = \Phi$. For the sake of simplicity and without a loss of generality, we impose that power states are equally spaced in terms of network performance:

$$\mu^{(s)} = S^{-1}(s+1) \tag{4}$$

and then $\mu^{(0)} = S^{-1}, \mu^{(1)} = 2S^{-1}, \ldots, \mu^{(S-1)} = 1$.

Starting from the previous considerations, to estimate the maximum energy consumption in the sth power state, we apply the following relationship:

$$\Phi_a^{(s)} = \left(\Phi_{a\min}^{\frac{1}{\nu}} + \left(\Phi_{a\max}^{\frac{1}{\nu}} - \Phi_{a\min}^{\frac{1}{\nu}}\right)\mu^{(s)}\right)^{\nu} \tag{5}$$

It is worth noting that, when $\nu = 1$, we will have a linear relationship between the service capacity and the energy consumption, which is a typical condition of DFS approaches in many Component Off-The-Shelf (COTS) processors [49], while for $\nu = 2$, we can model the typical quadratic behaviour of DVS approaches in CMOS silicon. Higher values of ν could be suitable for representing DVFS approaches. Similar concepts and formulations were used also in [40].

3.2. The LPI model

As outlined in previous works [24–26,34], the average energy consumption of a generic device with LPI primitives enabled depends on the incoming traffic volumes with respect to the system's maximum capacity, and on the HW efficiency in entering and exiting idle modes. To represent such aspects, we modelled the average energy consumption $\widetilde{\Phi}^{(s)}$ of a device, including its idle logic and power scaling capabilities, as follows:

$$\widetilde{\Phi}^{(s)}(\lambda) = \left([\Phi_i^{(s)}]^{\frac{1}{\sigma}}\left(1 - \frac{\lambda}{\mu^{(s)}}\right) + \left[\Phi_a^{(s)}\right]^{\frac{1}{\sigma}}\frac{\lambda}{\mu^{(s)}}\right)^{\sigma} \tag{6}$$

where $\Phi_i^{(s)}$ is the power consumption during idle periods and σ is the shape parameter of the power consumption curve (we recall that $\lambda/\mu^{(s)}$ and $1 - \lambda/\mu^{(s)}$ represent the fractions of time spent in active and idle periods, respectively). In detail, the case $\sigma = 1$ corresponds to the ideal case, where there is no overhead in entering and exiting from idle states and the energy consumption increases linearly with respect to the traffic load λ. As previously sketched, we expect $\Phi^{(s)}$ to increase in a convex way with respect to λ given the non-negligible idle-active transition costs. Therefore, we assume $\sigma < 1$ for real implementations of idle logic [25].

Moreover, $\Phi_i^{(s)}$ may also depend on the selected power-state:

$$\Phi_i^{(s)} = \gamma\left(\Phi_a^{(s)} - \Phi_{i\min}\right) + \Phi_{i\min} \tag{7}$$

where γ is the dependency degree of idle logic on the power states: if $\gamma = 0$ the energy consumption in idle periods does not depend on the sth state. Conversely, if $\gamma = 1$, we have no idle logic optimisations in the considered device.

In more detail and as previously mentioned, to provide a conservative estimation of energy saving, we decided to impose $\Phi_{i\min} = \Phi_{a\min}$.

3.3. Using the optimal energy profile with AR and/or LPI primitives

A device providing more than one power state should be controlled by optimisation frameworks working on the single device or among multiple network nodes. The specific objective of such frameworks is to select the best power states by providing the minimum energy consumption, while meeting the incoming traffic load and Quality of Service (QoS) requirements.

From a very simplistic point of view, the output of such an optimisation can be synthesised as follows:

$$\widetilde{\Phi}(\lambda) = \min_{s=0\ldots S-1}\left\{\widetilde{\Phi}^{(s)}(\lambda) | \lambda(1+\xi) \leqslant \mu^{(s)}\right\} \tag{8}$$

where $\xi > 0$ represents how much the optimisation framework is "conservative" in allocating resources in terms of choosing the most suitable power state for meeting the

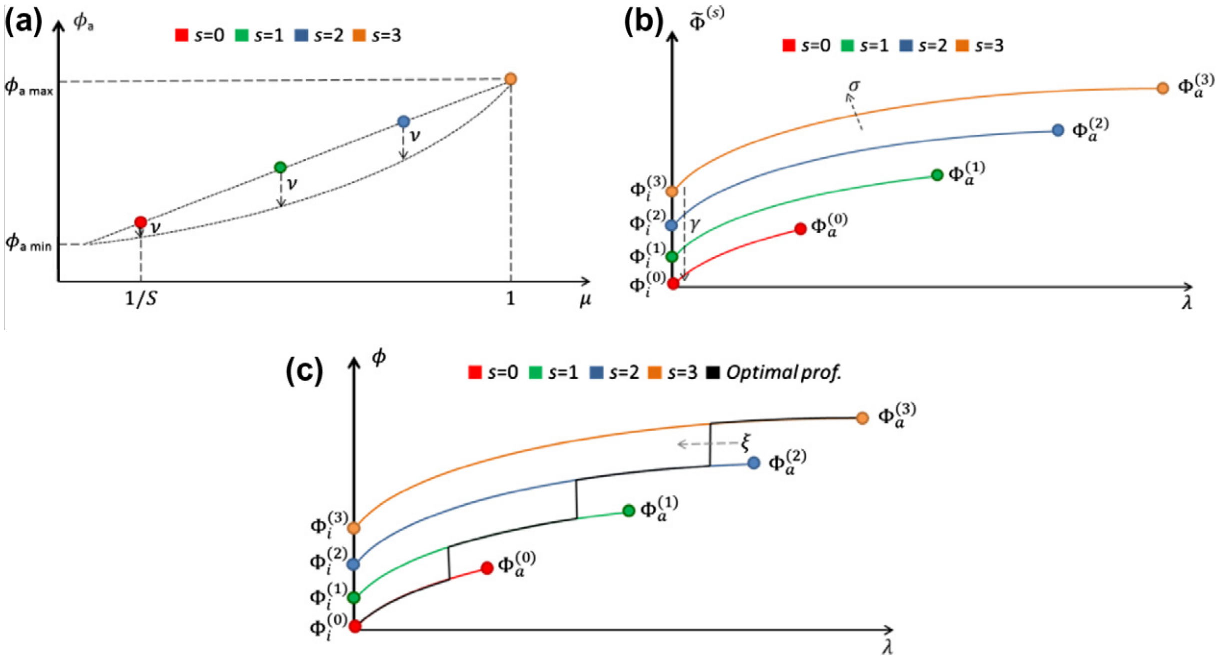


Fig. 3. An example of energy profiles in various AR states in the presence of LPI primitives and the resulting optimal device energy profile. (a) Shows the main impact of ν on $\Phi_a^{(s)}$ values according to Eq. (5), (b) shows the impact of γ and σ on $\widetilde{\Phi}^{(s)}$ according to Eq. (6). Finally, (c) depicts the main role of ξ on the optimal selection of $\widehat{\Phi}(\lambda)$ (see Eq. (8)).

incoming traffic load with the desired QoS level. To make the proposed model as intuitive as possible, Fig. 3 shows an example of the energy profile according to each AR state (obtained from Eq. (6)) and the final optimal one (obtained from Eq. (8)) for a generic device. The number of AR profile curves is obviously equal to the number of available states (S).

The optimal profile simply consists of a piece-wise curve, composed of the most power saving parts of AR curves of the available states guaranteeing a certain bound on link utilisations: $\rho^{(s)} = \lambda/\mu^{(s)} \leqslant 1/(1+\xi)$. As outlined in Fig. 3, by increasing ξ, the optimal energy profile would "jump" from the AR state with lower performance to the higher one for lower values of λ. The parameter ν affects how the maximum values of the AR curves are spaced among themselves. In the example in Fig. 3, ν is fixed at 1, and the curve joining all the maximum consumption levels of AR state profiles (i.e., $\Phi_a^{(s)}$) corresponds to a straight line; moreover, $\Phi_a^{(s)}$ values are then equally spaced. When $\nu > 1$, this curve becomes more convex and the $\Phi_a^{(s)}$ values tend to become more spaced according to the increase in s. In an analogous way, γ impacts the energy consumption of the AR state profiles when $\lambda = 0$. If γ is fixed at 0, all of the AR state profiles would start from the same level of energy consumption. Finally, by increasing σ above 1, the curve of the AR state profiles would become more concave. When $\sigma = 1$, the profiles become straight lines between $\Phi_a^{(s)}$ and $\Phi_i^{(s)}$.

3.4. Standby model

The energy consumption of the entire device in standby state in the presence of standby/sleeping primitives is obtained using Eq. (3).

Actually, we consider that standby/sleeping primitives may be applied to entire devices or, depending on the prototype architecture, to selected parts of them (e.g., to a single line card, or to a link interface, etc.). However, the model proposed here does not explicitly address the energy-profile of each sub-component, but only provides an aggregated representation. We consider putting into standby status the unused parts of the network in terms of nodes and links, which do not cause a potential "connectivity loss". For instance, as under investigation inside the ECONET project, we consider putting into standby mode all the redundant hardware (in terms of entire backup nodes or backup interfaces/line cards) deployed in the network for resilience purposes. In addition to redundant HW, we also consider the effect of energy-aware traffic engineering and routing policies, which may allow aggregating traffic in a subset of nodes and links during low traffic volume periods, and consequently put in standby status further HW across the network.

Thus, as shown also later in Section 5, our approach leads to the application of standby primitives, especially on metro, transport and core network segments, where redundant HW elements and more alternative paths are generally present. At the (home) customer access level, we only use LPI primitives.

3.5. Considering the entire network infrastructure

Similar to the work in [17], we consider a network infrastructure composed of four network segments: home, access, metro/transport and core. The representative devices for such network levels are home gateways, Digital Subscriber Line Access Multiplexers (DSLAMs), high-end optical switches and multi-chassis IP routers. Thus, to represent such a network structure in a compact way, we will use vector notation. The vectors and matrixes will be denoted in bold style.

We start by introducing vectors $\mathbf{\Gamma}$ and $\mathbf{\Omega}$, which have four elements representing the device density and related BAU power consumption (Φ) for the home, access, metro/transport and core networks, respectively. Thus, the BAU energy requirement of the entire network infrastructure can be easily expressed as:

$$E_{BAU} = \mathbf{\Gamma} \cdot \mathbf{\Omega} \tag{9}$$

where the dot represents the scalar product.

To estimate the energy gain of green technologies, we need to estimate which share of HW (in terms of entire devices or parts of them) can enter standby mode and, consequently, how much HW is active and working with LPI and AR.

Thus, we can simply obtain the share of redundant HW (to be put in standby mode) by considering both the share of fully redundant devices and the redundant links on "active" devices as follows:

$$\boldsymbol{\theta_{red}} = \frac{\boldsymbol{\theta_d}}{2} + \left(1 - \frac{\boldsymbol{\theta_d}}{2}\right)\mathbf{I}_4\frac{\boldsymbol{\theta_l}}{2} \tag{10}$$

where $\mathbf{I}_4$ is the identity matrix of rank 4, while $\boldsymbol{\theta_d}$ and $\boldsymbol{\theta_l}$ are the average redundancy degrees of devices and links, respectively.

As previously sketched, the adoption of green traffic engineering and routing policies may help in aggregating the transport traffic flows in a subset of nodes and links, especially during night hours with low traffic volumes. Therefore, additional HW (entire devices and links) can be emptied by traffic and can enter standby modes. We estimate the additional share of HW entering standby modes due to green traffic engineering as follows:

$$\boldsymbol{\theta_\alpha} = \boldsymbol{\theta_{red}} + (1 - \boldsymbol{\theta_{red}})\mathbf{I}_4\boldsymbol{\delta} \tag{11}$$

However, aggregating the traffic in few nodes and links causes an increase in the traffic load on the ones remaining active. The increase factor of traffic volumes in devices remaining active can be expressed as the ratio of active network resources without such policies to the number of active ones with the policies:

$$\boldsymbol{\omega}^{(j)} = \frac{1 - \boldsymbol{\theta}^{(j)}_{\boldsymbol{red}}}{1 - \boldsymbol{\theta}^{(j)}_{\boldsymbol{\alpha}}} \quad \text{with } j = 0, \ldots, 3 \tag{12}$$

where j represents the jth element of $\boldsymbol{\omega}$, $\boldsymbol{\theta_{red}}$ and $\boldsymbol{\theta_\alpha}$ vectors.

Extending Eq. (9), the energy consumption of the entire network infrastructure with green technologies (GT) enabled can be easily expressed as the energy consumption of sleeping HW elements and active ones:

$$E_{GT} = \left[\boldsymbol{\theta_\alpha}\mathbf{I}_4\Phi_s + (1 - \boldsymbol{\theta_\alpha})\mathbf{I}_4\widetilde{\boldsymbol{\Phi}}_{\boldsymbol{real}}\right] \cdot \mathbf{\Gamma} \tag{13}$$

where $\widetilde{\boldsymbol{\Phi}}_{\boldsymbol{real}}$ is the average power consumption (per network segment) of active devices working with AR and LPI in the presence of fluctuating incoming traffic loads.

In more detail, to capture the effect of daily traffic profiles shown in Section 4 below and recalling Eq. (8), we express $\widetilde{\boldsymbol{\Phi}}_{\boldsymbol{real}}$ as the expectation of $\widetilde{\Phi}(\lambda\boldsymbol{\omega})$ with respect to the incoming traffic volume λ and the increase of traffic $\boldsymbol{\omega}$ due to the additional resources put in standby mode by the network control policies:

$$\widetilde{\boldsymbol{\Phi}}_{\boldsymbol{real}} = \int_0^1 \widetilde{\Phi}(\lambda\boldsymbol{\omega})p_\lambda(\lambda)d\lambda \tag{14}$$

Finally, the energy savings with respect to the BAU case can be easily found as follows:

$$\Delta E_{GT} = E_{BAU} - E_{GT} \tag{15}$$

4. Reference network scenarios

In the reference network scenarios, the goal of this paper is to quantitatively evaluate the potential gain from the application of green networking technologies. The trade-off among the device performance and the energy savings upon the application of energy-aware technologies is examined. To make our impact analysis as realistic as possible, we consider the energy efficiency targets of the ECONET project [50] and apply them to two scenarios. The first scenario is based on the network of Telecom Italia – a large-scale Telco; the second scenario is based on the National Research and Education Network (NREN) of GRNET (Greek Research and Technology Network), which can be considered as a medium size ISP.

4.1. Telecom Italia reference scenario

The first reference scenario, as also presented in [17], is the next generation network of Telecom Italia (TELIT). The TELIT network is structured on 32 Points of Presence (PoPs) to concentrate all data access technologies and is specifically dimensioned to have a peak utilisation equal to 50% of its allowable bandwidth in each link, except at the link terminations. It is worth noting that 50% is only the maximum allowable link utilisation; the real utilisations are often much lower. The current situation and the historical data for the energy requirements of the TELIT network infrastructure were already mentioned in the introduction of the paper. The objective of analysing this scenario is to move a step beyond these data to evaluate how these energy figures will change in the near future. One of the most likely evolutions of the TELIT network, which is going to be deployed by 2015–2020, is considered.

This evolution refers to a network with 17.5 million customers, in which the presence of broadband-only access technologies is assumed, together with suitable over-provisioning in the metro, transport and core segments. The example reference scenario is shown in Table 2, which contains the number of devices per network segment and their energy consumption in the Business-As-Usual case (i.e., the case in which no green enhancements are included in network devices). Both the end-user and the operator equipment have been taken into account. The devices' energy consumption has been forecasted based on current values and high quality specifications (e.g., European Broadband Code of Conduct) and the expected "inertial" technological improvements (e.g., Dennard's law [16]). The devices' energy requirements include

Table 3
2015–2020 network forecast: device density and energy requirements in the business-as-usual case (BAU). Example based on the Telecom Italia network.

Network part	Power consumption (W)	Number of devices (#)	Overall consumption (GWh/year)
Home	10	17,500,000	1533
Access	1280	27,344	307
Metro/transport	6000	1750	92
Core	10,000	175	15
	Overall network consumption		1947

Table 4
Traffic and topological data (average figures) for the 2015–2020 perspective network. *Source*: Telecom Italia.

Network part	Network topological index	Value
Home access	number of customers per DSLAM	640
	link utilisation when a user is connected	10%
Core, transport and metro	redundancy degree for metro/transport devices	13%
	Redundancy degree for core devices	100%
	Redundancy degree of metro/transport links	100%
	Redundancy degree of core device links	50%
	Link utilisation in metro networks	40%
	Link utilisation in core networks	40%

the contribution of site cooling and powering systems, which account for an average 36% of a device's direct consumption.

Starting from the scenario in Table 3, the per-user average energy requirement consists of approximately 111 kWh per year mainly in home and access networks, accounting for 79% and 16% usage, respectively. Metro/transport and core networks account only for 5%, but their joint energy requirement of approximately 107 GWh per year (about a quarter of the Telco's direct energy consumption) can be a convincing driver for reducing the carbon footprint of backbone devices in the near future.

Table 4 defines the key parameters that allow the synthetic modelling of the average usage of network devices and links. The values have been derived by defining the expected (by 2015–2020) average customer up-times and loads, the average traffic utilisation on metro/transport and core networks and the number of devices and links that are usually deployed for redundancy purposes. It is worth noting that the traffic load values in Table 3 are significantly larger (and, consequently, give rise to conservative consumption estimations) than those of the current network.

Redundancy is distinguished in device level redundancy and network links redundancy. In the first case, redundancy can be supported through the existence of back-up devices that are activated in case of problems in the master devices or the existence of redundant modules or cards within the same device. In the second case, redundancy can be supported through the existence of multiple paths between two devices in the network. The paths may follow similar (two or more links are activated between two devices, e.g., in different ports) or alternative routes (alternative paths exist through different nodes in the network).

Table 5
Maximum power consumption for the access and core network devices in the GRNET network.

Network part	Number of devices	Type of equipment	Vendor/model	Maximum power consumption
Access	42	Switch	Cisco Catalyst 2970, Extreme X350, Extreme X450a	190 W, 75 W, 659 W
Core	11	Router	Juniper T1600, Juniper MX960, Cisco 12406, 12410, 12416	~8000 W, ~3000 W, 5450 W

Table 6
Topological data (average figures) for the GRNET network. *Source*: Greek Research and Technology Network.

Network part	Redundancy degree type	Redundancy degree value (%)
Access	Redundancy degree for access devices	13
	Redundancy degree of access links	88
Core	Redundancy degree for core devices	72
	Redundancy degree of core links	71

4.2. Greek Research and Technology Network reference scenario

The example reference scenario by GRNET refers to devices in the core and access network. GRNET operates a next-generation optical fibre network based on Wavelength Division Multiplexing – WDM technology at high speeds (1–10 Gbps). The core network is formed by IP routers that are interconnected with 2.5 Gbps circuits over 10 Gbps wavelengths implemented via proprietary DWDM equipment. The GRNET network can be divided into core and access parts. The access network consists of dark fibre pairs between the PoP of GRNET in each major city in Greece and the PoP of the connected university or research institute. Approximately 100 clients are connected to the GRNET network, whose topology [51] can be considered as a flat one without large aggregation points. Alternative backup paths are available for interconnecting the majority of the network nodes, while more than one alternative path exists for interconnecting the core network nodes.

The IP/Ethernet network consists of 11 routers and roughly 42 switches from different vendors. The typical energy consumption of each network device differs according to the installed cards and modules. The maximum power

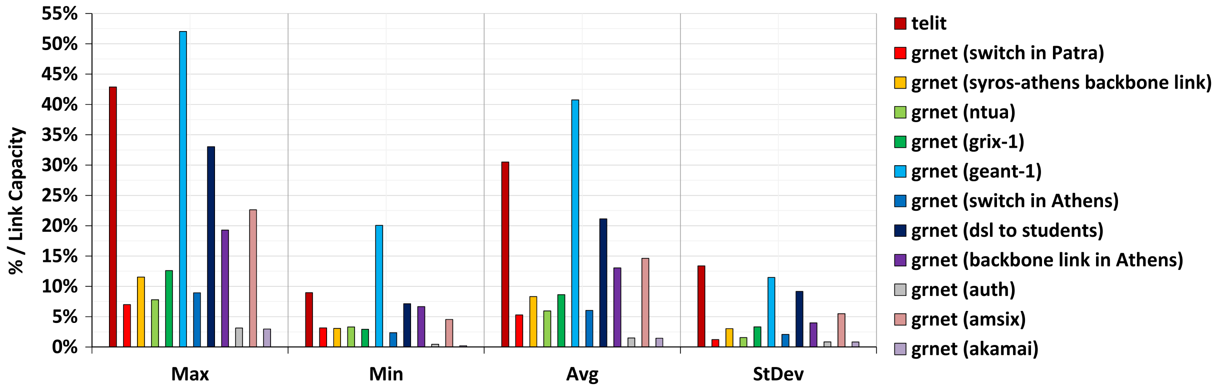


Fig. 4. The maximum, minimum, average and standard deviation with respect to link capacity for all of the network links during working days.

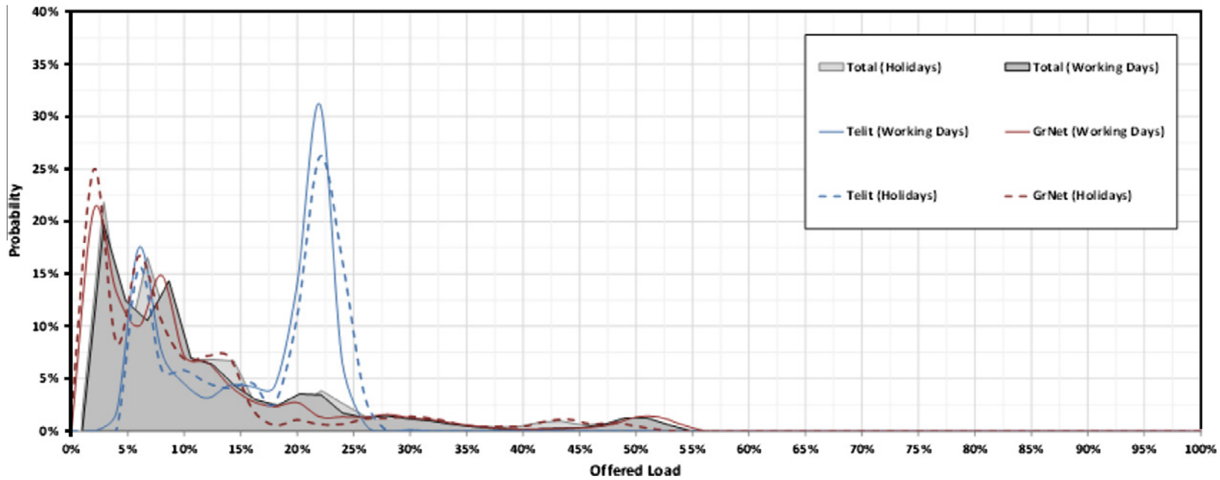


Fig. 5. The aggregated probability distribution of traffic loads in all of the network links during working days and holidays.

consumption for the access network switches and the core network routers, when fully loaded in terms of cards, modules and overall utilisation, is shown in Table 5. The DWDM equipment employed in the core has a total maximum power consumption of 18,320 Wh. The power consumption of each core router has been recorded in real time for a period of 3 weeks [50]. The collected results indicate almost constant values in power consumption for all the routers, with small variations in some cases. However, these variations cannot be related to the amount of traffic served from a router, but are most likely due to starting and stopping the device's cooling fans. Table 6 introduces the key parameters that contribute to the average usage of network devices and links based on the current version of the GRNET network. The redundancy degree of core network devices is below 100% because many routers do not support redundancy in the routing engine; for the access network devices, redundancy in the power supply is not always supported. Furthermore, redundancy in access network links is low because, in most cases, these links correspond to the interconnection of GRNET clients – mainly local universities and research institutes – with the closest PoP, and there is a low probability of network link failures.

4.3. Traffic profiles

In addition to the description of the network topologies and the devices of the considered reference networks, a detailed profile of their current traffic is required to consider the potential of deploying energy-efficient strategies. Traffic data for representative links in the TELIT and GRNET network has been collected and the basic characteristics of the extracted traffic profiles are presented. The main result of this analysis is that the network links appear to be under-utilised and that the maximum link occupancy is lower than half of the link capacities in most cases. These results were expected because, as already mentioned, network infrastructures and links are generally dimensioned for at least twice the carried traffic volumes. Fig. 4 compares the maximum, minimum, average and standard deviation statistics with respect to the link capacity for various links in the TELIT and GRNET networks during working days. These statistics represent a normalised value of link utilisation, so that they can be compared with each other. The maximum traffic load is less than 55% of the link capacity for every considered network link and is higher than 25% only in a small part of the network links. Hence, the link utilisation is never at maximum capacity,

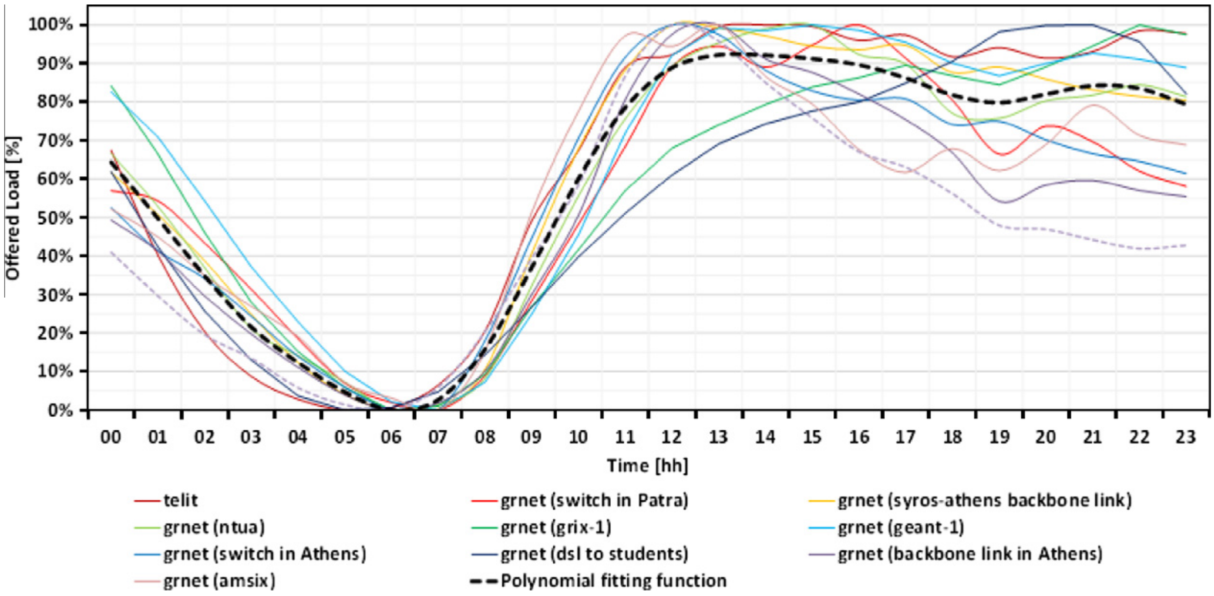


Fig. 6. The normalised traffic loads in all of the network links during working days.

which provides the opportunity to implement energy saving policies according to network traffic. To individuate the link utilisation in the considered network links, Fig. 5 shows the probability to have a given level of offered load (with respect to the link capacity) for the working day profile and the holiday profile. The highest probability is for the range between 20% and 25% link utilisation for the TELIT network links and for the range between 5% and 10% for the GRNET network links, followed by the one between 5% and 10%. Note that the differences between the two profiles (working day and holiday) are not significant and these considerations are common to both cases. Because the probability that the offered load exceeds 30% of the link capacity is very low, it is likely that significant energy savings can be obtained by setting the speed of the links based on the actual network traffic.

As shown in Fig. 6, the evolution of the traffic offered load has the classical night-and-day profiles. The minimum of the traffic is noticed during the first hours of the morning (from 03:00 to 06:00), while rush hours are during the day. A similar trend is also noticed in case of the holiday profile, as shown in Fig. 7.

5. Estimating the energy saving gains

This section evaluates the potential impact estimated by the proposed model within the energy-efficiency degrees suggested by the ECONET consortium for realistic scenarios provided by Telecom Italia and GRNET for the access, home, metro/transport and core networks. The main target of this analysis is to understand the impact of each energy adaptation primitive on the various network segments and to consequently collect precious information on how these technologies have to be designed and which level of efficiency they should provide for maximally reducing the energy consumption in real-world ISP and Telco network infrastructures. As far as the network scenarios are concerned, we fixed the vectors $\mathbf{\Gamma}$ and $\mathbf{\Omega}$ for GRNET and Telecom Italia by starting from Tables 3 and 5:

$$\mathbf{\Gamma}_{TELIT} = \begin{bmatrix} 17500000 \\ 27344 \\ 1750 \\ 175 \end{bmatrix} \quad \mathbf{\Omega}_{TELIT} = \begin{bmatrix} 10 \\ 1280 \\ 6000 \\ 10000 \end{bmatrix}$$
$$\mathbf{\Gamma}_{GRNET} = \begin{bmatrix} 0 \\ 42 \\ 0 \\ 11 \end{bmatrix} \quad \mathbf{\Omega}_{GRNET} = \begin{bmatrix} 0 \\ 297 \\ 0 \\ 3677 \end{bmatrix} \tag{16}$$

It is worth noting that the network access segment of Telecom Italia (the 2nd row's elements of the vectors above) has a quite different nature in terms of its device typologies and technologies from the GRNET one because the former network corresponds to the residential access, whereas the latter corresponds to peering to local academic networks, research institutes or data-centres. In terms of device typology (mainly L2 switches), the access segment of GRNET roughly corresponds to the metro segment of Telecom Italia.

Starting from the values in Tables 4 and 6, we can define the redundancy degrees per network level in both of the considered scenarios:

$$\boldsymbol{\theta}_{\mathbf{d}|\mathbf{TELIT}} = \begin{bmatrix} 0.00 \\ 0.00 \\ 0.13 \\ 1.00 \end{bmatrix} \quad \boldsymbol{\theta}_{\mathbf{d}|\mathbf{GRNET}} = \begin{bmatrix} 0.00 \\ 0.13 \\ 0.00 \\ 0.72 \end{bmatrix}$$
$$\boldsymbol{\theta}_{\mathbf{l}|\mathbf{TELIT}} = \begin{bmatrix} 0.00 \\ 0.00 \\ 1.00 \\ 0.50 \end{bmatrix} \quad \boldsymbol{\theta}_{\mathbf{l}|\mathbf{GRNET}} = \begin{bmatrix} 0.00 \\ 0.88 \\ 0.00 \\ 0.71 \end{bmatrix} \tag{17}$$

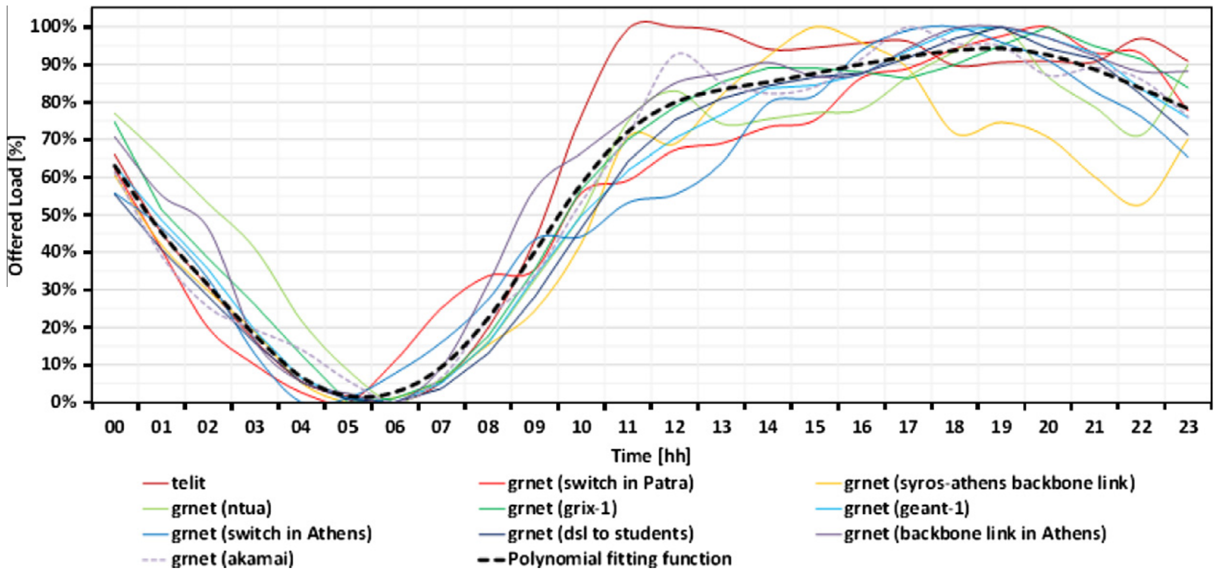


Fig. 7. The normalised traffic loads in all of the considered network links during holidays.

We used the data shown in Section 4.3 obtained by real measurements in the GRNET and Telecom Italia networks for the traffic profiles. In more detail, the traffic load measurements were used to estimate the load probability distribution $p_\lambda(\lambda)$, as shown in Fig. 6, while, to make a conservative estimation, the maximum link occupancy was fixed at 50% of the available capacity.

As far as the devices in the various network segments are concerned, we use the same energy consumption breakdown as in [17], coming from device manufacturers in the ECONET consortium:

$$\psi_{ctr} = \begin{bmatrix} 0.03 \\ 0.03 \\ 0.13 \\ 0.11 \end{bmatrix} \quad \psi_{data} = \begin{bmatrix} 0.95 \\ 0.80 \\ 0.73 \\ 0.54 \end{bmatrix} \quad \psi_{cool} = \begin{bmatrix} 0.02 \\ 0.17 \\ 0.14 \\ 0.35 \end{bmatrix} \tag{18}$$

In a similar way, we used the energy efficiency values indicated by the ECONET consortium:

$$\boldsymbol{\alpha} = \begin{bmatrix} 0 \\ 0.8 \\ 0.8 \\ 0.8 \end{bmatrix} \quad \beta = \begin{bmatrix} 0.7 \\ 0.65 \\ 0.65 \\ 0.75 \end{bmatrix} \quad \boldsymbol{\kappa} = \begin{bmatrix} 0.15 \\ 0.15 \\ 0.15 \\ 0.15 \end{bmatrix}$$
$$\boldsymbol{\gamma} = \begin{bmatrix} 0 \\ 0.05 \\ 0.01 \\ 0.01 \end{bmatrix} \quad \boldsymbol{S} = \begin{bmatrix} 5 \\ 5 \\ 6 \\ 5 \end{bmatrix} \quad \boldsymbol{\sigma} = \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}$$
$$\boldsymbol{\delta} = \begin{bmatrix} 0 \\ 0 \\ 0.2 \\ 0.2 \end{bmatrix} \quad \boldsymbol{\nu} = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix} \quad \xi = \begin{bmatrix} 0.1 \\ 0.1 \\ 0.1 \\ 0.1 \end{bmatrix} \tag{19}$$

The rest of this section is organised with the following aim. Section 5.1 estimates the energy-saving that green technologies can provide for the Telecom Italia and GRNET network infrastructures if implemented according to the efficiency parameters defined in the ECONET project. Then, this outcome is compared with the "Business As Usual" scenarios to understand the benefits for each network segment and each green primitive under investigation. Using these results as a fixed point, in Section 5.2, we then estimate the impact on the energy savings gains by varying different parameters of the proposed energy profile model. In this analysis, each method will be investigated singularly to identify the behaviour of the proposed energy profile model.

5.1. The overall gains with the green technologies from the ECONET project

The yearly energy consumption in the BAU scenario is estimated by Eq. (9) using the energy consumption and the number of devices of the two networks in Eq. (16). Fig. 8 shows the yearly energy consumption for the BAU scenarios decomposed per network segment. Most of the energy consumption for the Telecom Italia reference scenario is in the home network. In the GRNET reference scenario, as one can expect, most of the energy consumption is in the core segment.

Using Eq. (13), we compute the estimated energy consumption achievable by adopting the energy-efficiency parameters in Eq. (19) and the ones regarding device power breakdown and network topology of Eqs. (18) and (19). Fig. 9 shows the energy savings gains between the energy consumption estimated by the model including the DPS and stand-by primitives and the BAU scenario. From the results of these figures we can outline how the energy gains exhibit negligible differences between the working day and a holiday. In the Telecom Italia scenario, the largest part of the percentage savings is in the metro/transport and core networks, while the minimum is in the access network. Instead, for GRNET, the access network also has a high value of energy savings gain (in this case, we need

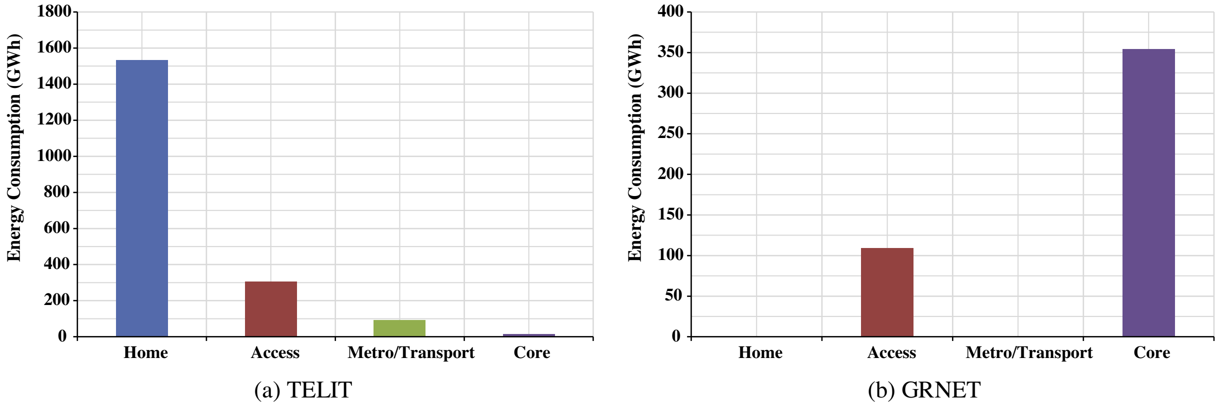


Fig. 8. The yearly energy consumption estimation for TELIT (a) and GRNET (b) home, access, metro/transport and core networks in the BAU scenario.

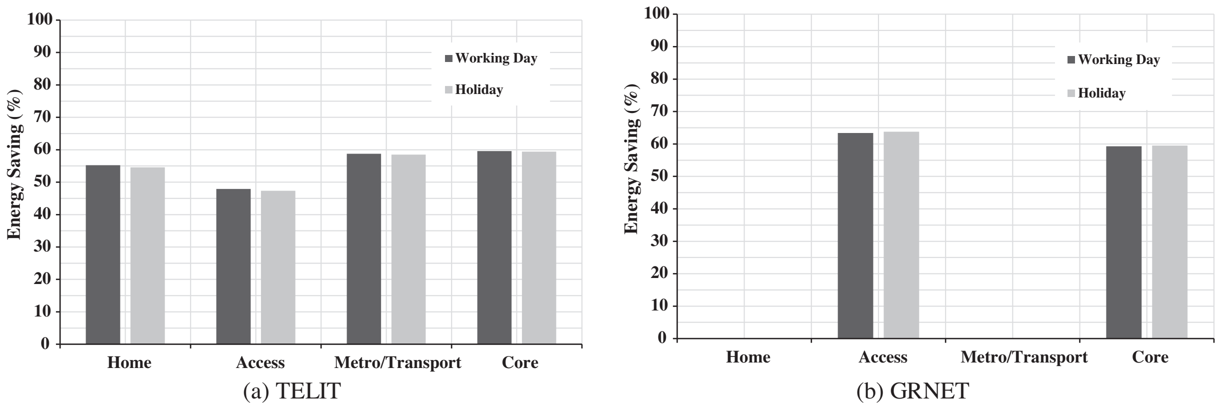


Fig. 9. The estimated energy savings for TELIT (a) and GRNET (b) home, access, metro/transport and core networks during the working day and holiday profiles between the proposed energy model and the BAU scenarios.

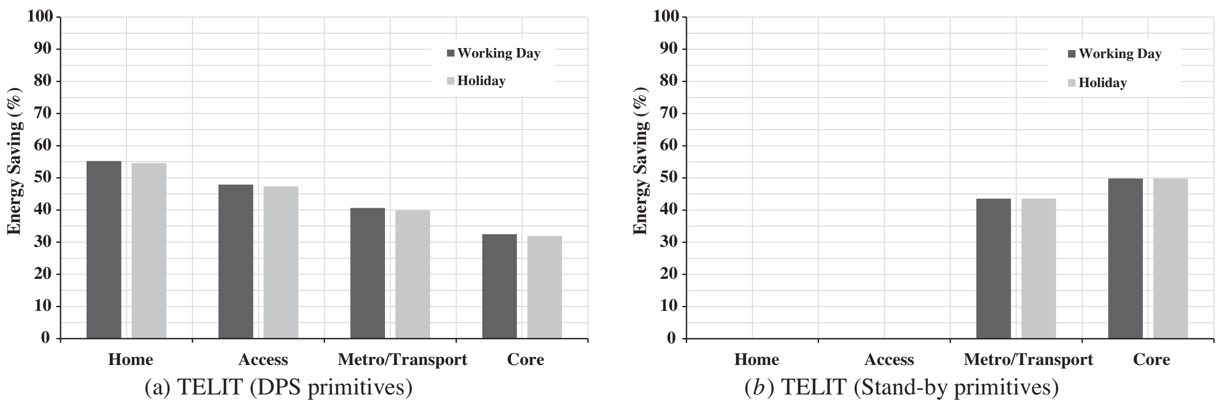


Fig. 10. The estimated energy savings for TELIT home, access, metro/transport and core networks during the working day and holiday profiles between the proposed energy model with only DPS (a) or stand-by (b) primitives and the BAU scenarios.

to recall that the access network of GRNET roughly corresponds to the metro/transport segment of a large-scale telecom operator).

To selectively evaluate the benefits from the various energy-aware primitives, we also estimated the energy savings gain achievable by enabling only DPS primitives or stand-by ones. Figs. 10 and 11 report these gains for Telecom Italia and GRNET traffic traces, respectively.

It is interesting to note that, for the Telecom Italia reference scenario, the DPS primitives contribute to the energy

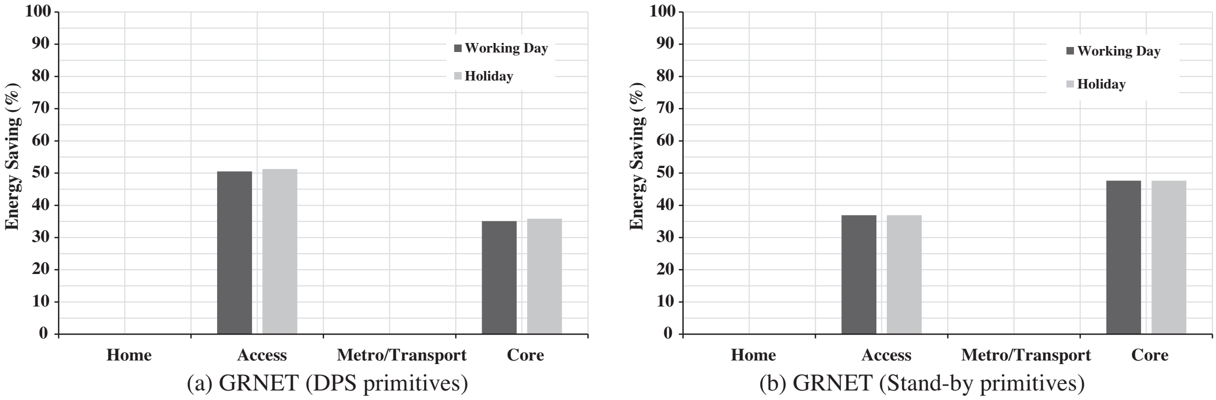


Fig. 11. The estimated energy savings for GRNET's home, access, metro/transport and core networks during the working day and holiday profiles between the proposed energy model with only DPS (a) or Stand-by (b) primitives and the BAU scenarios.

savings gain in every part of the network, while the stand-by ones yield a significant energy savings gain only for the metro/transport and core networks. This effect is simply because we assume the standby capabilities to be applied only where "alternative paths" are present and, even then, only to those segments with partially/fully meshed topologies. Obviously, residential access generally has no such features.

As a result, standby primitives seem to assure major gains in network segments with high meshing degrees, and DPS ones in the other cases.

5.2. Parameter evaluation

To better understand and characterise the impact of the specific design of the considered power management primitives, we decided to perform a deeper analysis of the energy savings gains according to variations of the model parameters, as described in Section 4 and reported in Table 7.

We consider the scenario of Section 5.1 as a reference and we study the impact of each parameter on the overall potential savings in the TELIT and GRNET networks within realistic traffic profiles. The main objective of this analysis is to understand the dependency of the gains on the efficiency degrees and shape parameter of the DPS and stand-by primitives. These results will provide crucial feedback on how these primitives have to be designed to assure the maximum gain.

This section is organised into multiple subsections, where the estimation of the energy consumption is computed according to each of the parameters shown in Table 7. Because the differences between the working days and holidays are not significant, as shown in the previous section, we consider only the working day case.

5.2.1. Evaluation of the parameter S

The first analysis consists of the evaluation of potential energy savings gains according to the number of available AR states (S). This parameter affects only the DPS primitives, so the results do not apply to stand-by aspects. We perform this evaluation by varying the number of available states from 2 up to 10.

Table 7
Energy model parameters considered in the evaluation.

S	Number of active states
ν	Shape of the power states' energy consumption
γ	Dependency degree of idle logic on the power states
σ	Shape of the idle optimisation efficiency
α	Stand-by efficiency
β	Performance scaling efficiency
κ	Air cooling/power supply efficiency
δ	Traffic engineering efficiency

Figs. 12 and 13 show the trend of energy savings gains with respect to the BAU scenario, considering the variation of S. As one can intuitively expect, the increase in the number of states increases the energy savings. However, most of the potential gain is achieved by only five available AR states, and the energy savings gains are quite constant for values of s higher than 10. Thus, the model suggests that a high number of AR states is not necessary because the energy savings gain cannot grow linearly with it. Actually, with a higher number of states, we also increase the complexity of the device's HW and the procedures for dynamically managing them.

5.2.2. Evaluation of the parameter ν

As described in Section 4.1, the ν parameter controls the DPS primitives and models the shape parameter, which represents how the trade-off between maximum performance and energy consumption changes according to the AR states.

We consider this parameter ranging from 0 to 10. When $\nu = 1$, we will have a linear relationship between the service capacity and the energy consumption, which is a typical condition of frequency scaling (DFS) approaches in many COTS processors, while for $\nu = 2$, we can model the typical quadratic behaviour of voltage scaling (DVS) approaches in CMOS silicon. Higher values of ν are suitable for representing dynamic voltage and frequency scaling (DVFS) approaches.

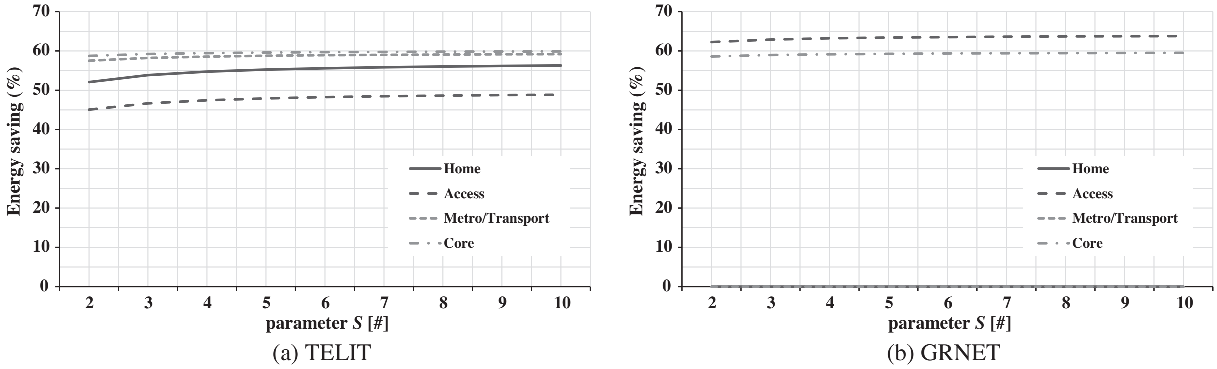


Fig. 12. The estimated energy savings for TELIT (a) and GRNET (b) home, access, metro/transport and core networks during the working day profile between the proposed energy model by varying the number of P-States (S parameter) and the BAU scenarios.

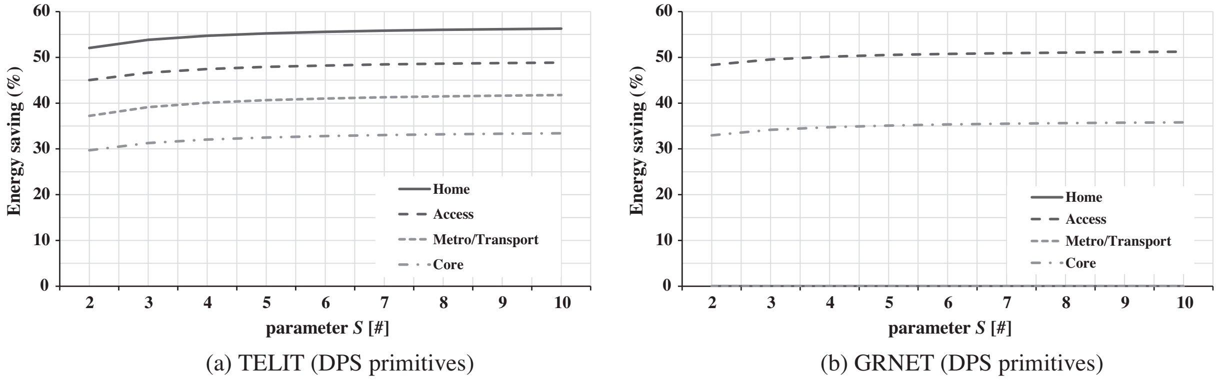


Fig. 13. The estimated energy savings for TELIT (a) and GRNET (b) home, access, metro/transport and core networks during the working day profile between the proposed energy model by varying the number of P-States (S parameter) with only DPS primitives and the BAU scenarios.

As shown in Figs. 14 and 15, the shape parameter does not significantly affect the energy savings; only for lower values $v \leqslant 2$, there is a slight improvement.

5.2.3. Evaluation of parameter γ

In the proposed model, the γ parameter represents the dependency degree of LPI performance (in terms of $\Phi_i^{(s)}$) on the selected AR state. Higher values of γ represent a heavy dependency of $\Phi_i^{(s)}$ on s. When $\gamma = 0$, there is no dependency of the $\Phi_i^{(s)}$ value on s. This determines a worsening in the energy savings gains with higher values of γ. The best results are with $\gamma = 0$.

Figs. 16 and 17 report the potential energy savings estimated by varying γ by considering all of the energy-aware primitives (DPS + standby) and only the DPS ones, respectively. Fig. 18 shows that, by only considering DPS primitives, the energy savings variation according to γ is the same in every part of the network and decreases according to γ.

This behaviour is because, for $\gamma = 0$, the power absorption of devices is constant according the AR states and is fixed to a minimum level equal to $\Phi_{i\ \min}$ (see Eq. (7)). Instead, when $\gamma > 0$ and the device works in AR states with high performance, $\Phi_i^{(s)} \gg \Phi_{i\min}$.

Moreover, it is worth noting that the total energy savings gain in Fig. 16a shows an overall figure for the Telecom Italia metro/transport and core networks similar to those derived from the standby primitives. Similar considerations are valid for the GRNET access and core network in Fig. 16b.

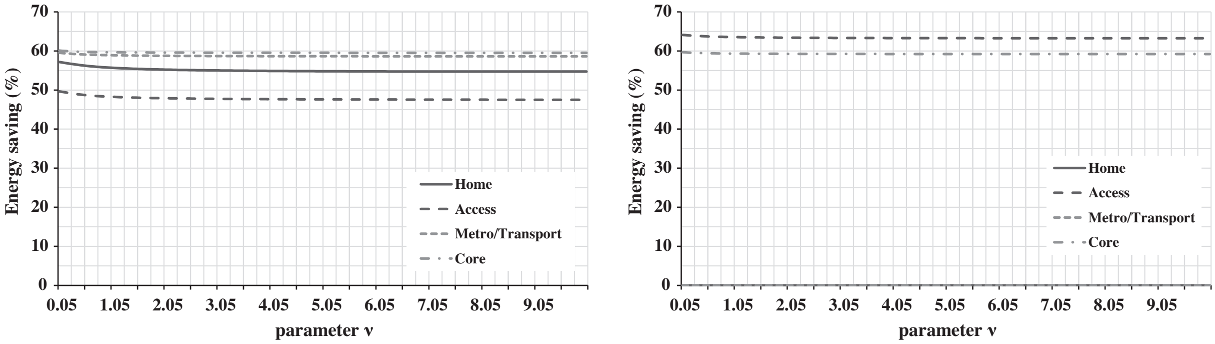
5.2.4. Evaluation of the parameter σ

The shape parameter of the LPI efficiency σ is taken to range from 0 to 1. In the ideal case ($\sigma = 1$), the power consumption with the DPS primitives is a linear combination of the LPI and AR mechanisms.

Fig. 18 (DPS + standby) and Fig. 19 (only DPS) show positive trends of energy gains. The best gain is obviously in the ideal case, but good performance can be achieved up to $\sigma = 0.5$. According to these results, it is worth noting that σ is a key parameter that should be used abundantly in green technology design because it can affect energy gains by approximately 15–20%, especially in telecom network access and home segments, where most of the energy is consumed.

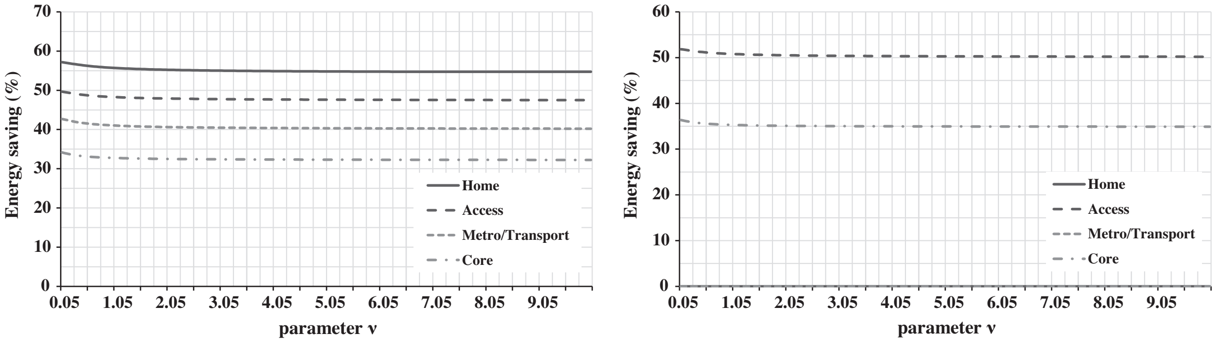
5.2.5. Evaluation of the parameter α

The standby efficiency α indicates how much energy can be saved when devices (or parts of them) enter the standby state. Obviously, by increasing this parameter,



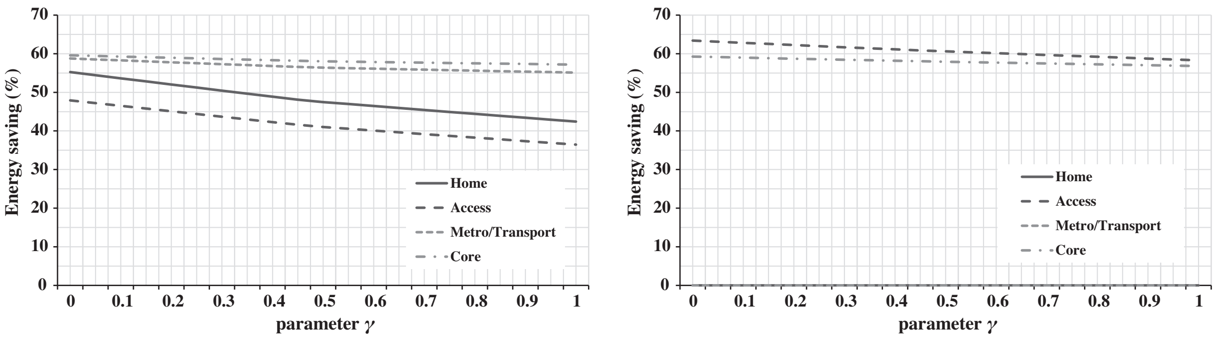
(a) TELIT (b) GRNET

Fig. 14. The estimated energy savings for TELIT (a) and GRNET (b) home, access, metro/transport and core networks during the working day profile between the proposed energy model by varying the shape of the power states' energy consumption (υ parameter) and the BAU scenarios.



(a) TELIT (DPS primitives) (b) GRNET (DPS primitives)

Fig. 15. The estimated energy savings for TELIT (a) and GRNET (b) home, access, metro/transport and core networks during the working day profile between the proposed energy model by varying the shape of the power states' energy consumption (υ parameter) with only DPS primitives and the BAU scenarios.



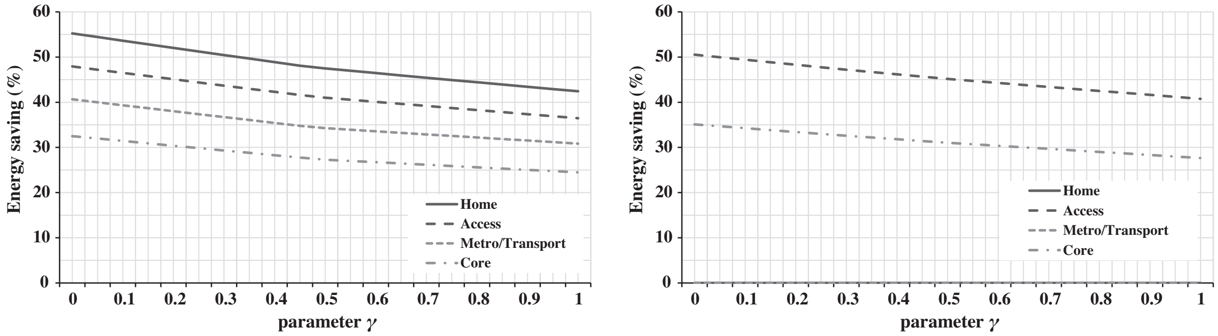
(a) TELIT (b) GRNET

Fig. 16. The estimated energy savings for TELIT (a) and GRNET (b) home, access, metro/transport and core networks during the working day profile between the proposed energy model by varying the dependency degree of idle logic on the power states (γ parameter) and the BAU scenarios.

there is a significant improvement in energy savings. However, the standby primitives affect energy gains only in such network segments where alternative paths exist (metro/transport and core networks for Telecom Italia, as well as the access and core segments of GRNET). Where standby primitives apply, Fig. 24 shows that α has a disruptive impact on energy savings because it allows savings of up to 45–60% of the total consumed energy. Moreover, the obtained results show a linear dependency between energy savings and α in both of the considered scenarios.

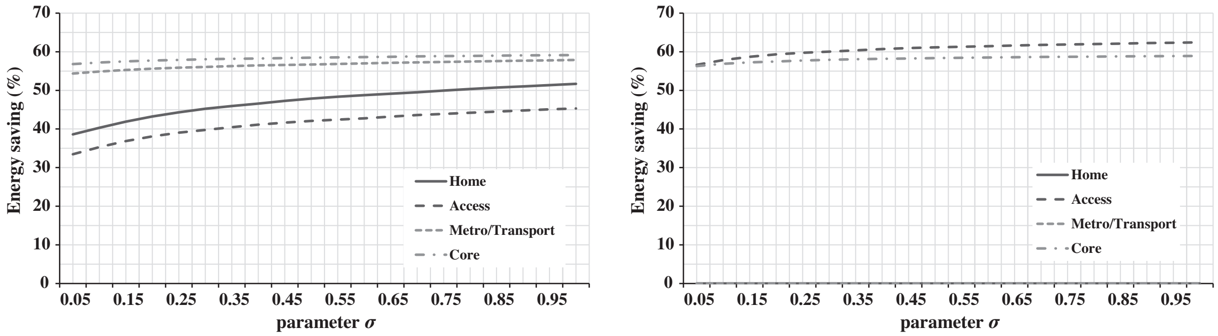
5.2.6. Evaluation of the parameter β

The parameter β represents the AR efficiency of a network device data-plane. When $\beta = 0$, AR is not enabled



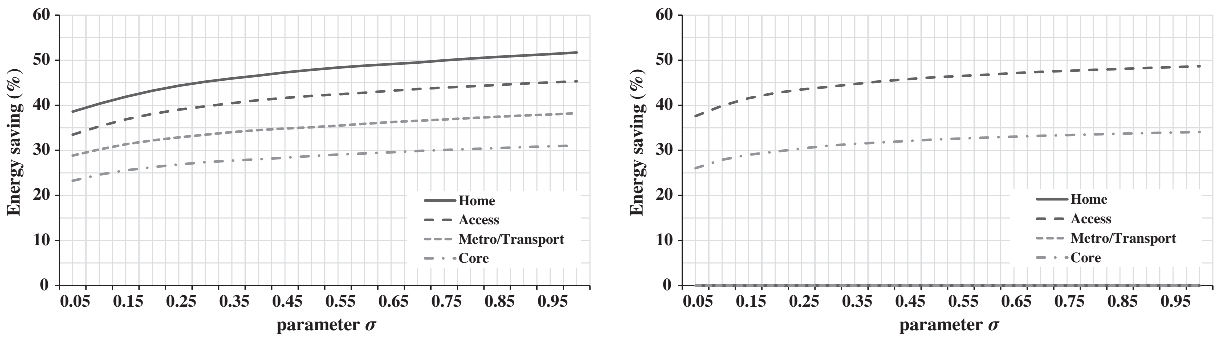
(a) TELIT (DPS primitives) (b) GRNET (DPS primitives)

Fig. 17. The estimated energy savings for TELIT (a) and GRNET (b) home, access, metro/transport and core networks during the working day profile between the proposed energy model by varying the dependency degree of idle logic on the power states (γ parameter) with only DPS primitives and the BAU scenarios.



(*a*) TELIT (*b*) GRNET

Fig. 18. The energy saving estimation for TELIT (a) and GRNET (b) home, access, metro/transport and core networks during working day profile between the proposed energy model by varying the shape of the idle optimisation efficiency (σ parameter) and the BAU scenarios.



(a) TELIT (DPS primitives) (b) GRNET (DPS primitives)

Fig. 19. The energy saving estimation for TELIT (a) and GRNET (b) home, access, metro/transport and core networks during the working day profile between the proposed energy model by varying the shape of the idle optimisation efficiency (σ parameter) with only DPS primitives and the BAU scenarios.

and then $\Phi_{amin} = \Phi$. Indeed, as shown in Figs. 25 and 26, the energy gains are very low for values of β close to 0, especially for the home and access networks of Telecom Italia. As previously mentioned, this behaviour is primarily due to the adoption of only DPS primitives within residential access and home networks.

In Fig. 25, in the presence of values close to 1, the best increment of the energy savings gain is in the home network (approximately 80%). The impact of β on the overall gain is quite similar to the one of the parameter α, but it is worth noting that β can positively affect every part of the network.

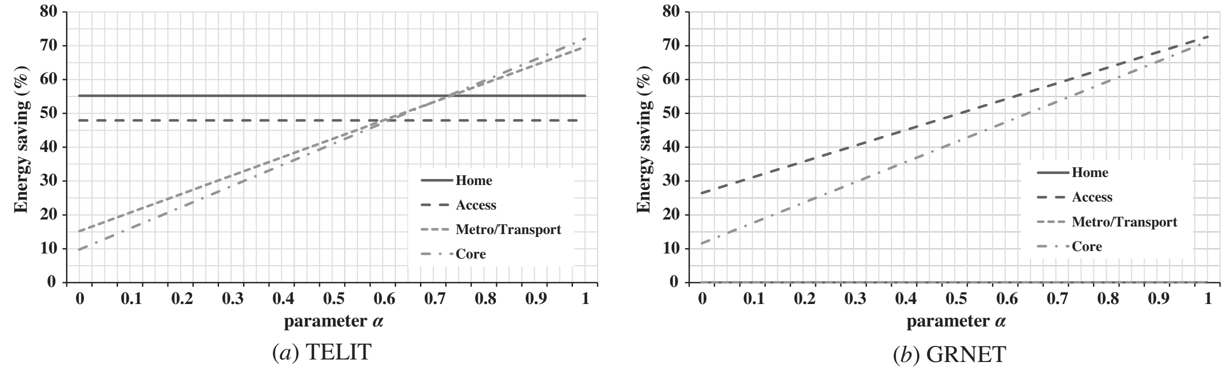


Fig. 20. The estimated energy savings for TELIT (a) and GRNET (b) home, access, metro/transport and core networks during working day profile between the proposed energy model by varying the Stand-by efficiency (α parameter) and the BAU scenarios.

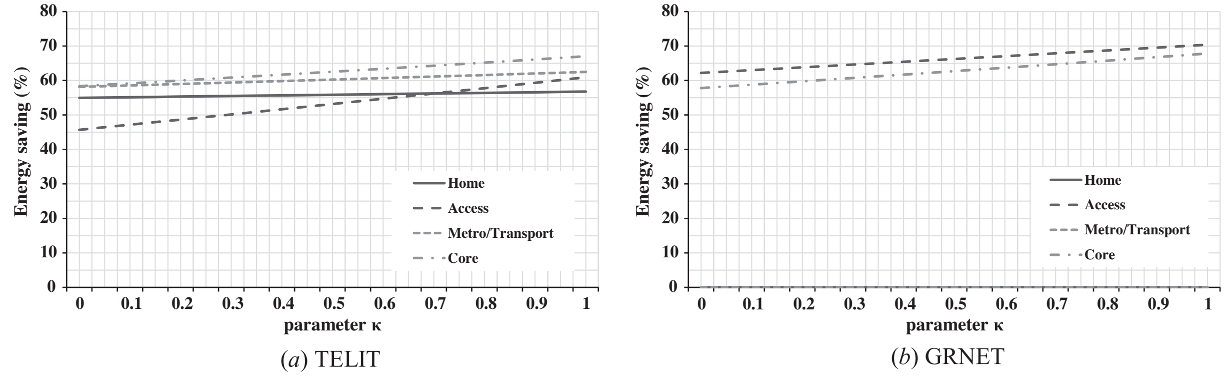


Fig. 21. The energy saving estimation for Telecom Italia (a) and GRNET (b) home, access, metro/transport and core networks during working day profile between the proposed energy model by varying the air cooling/power supply efficiency (κ parameter) with the BAU scenarios.

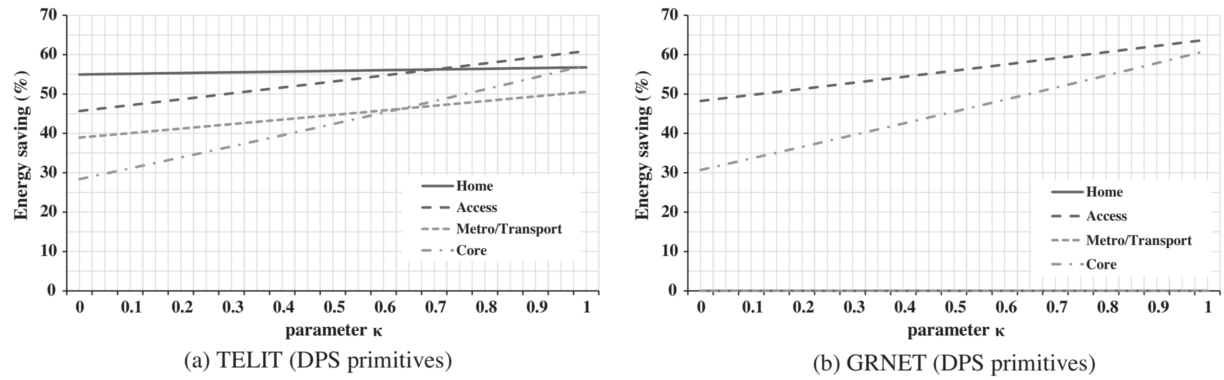
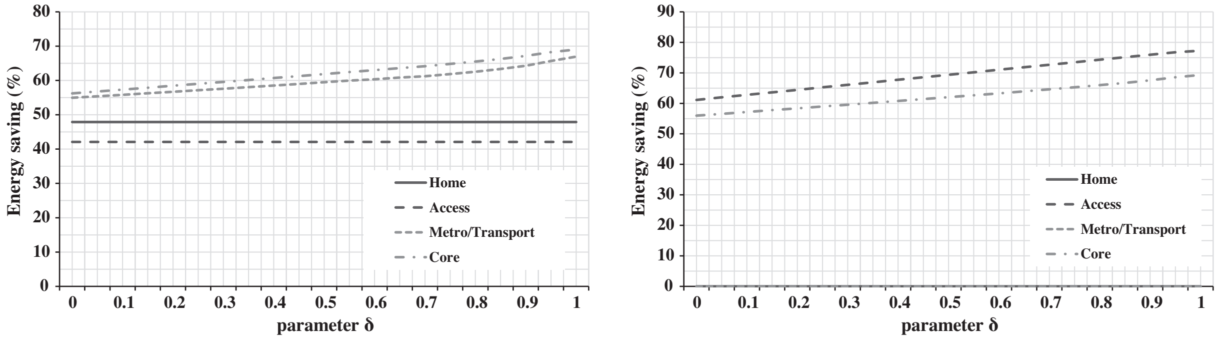


Fig. 22. The estimated energy savings for Telecom Italia (a) and GRNET (b) home, access, metro/transport and core networks during working day profile between the proposed energy model by varying the air cooling/power supply efficiency (κ parameter) with only DPS primitives and the BAU scenarios.

5.2.7. Evaluation of the parameter κ

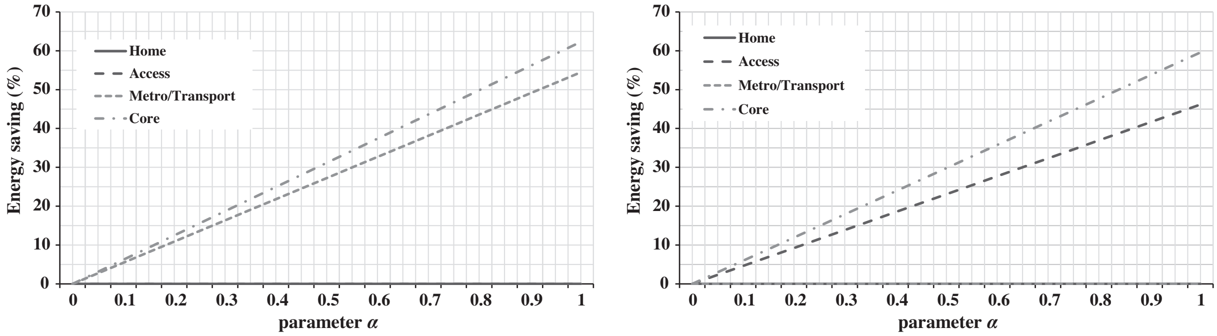
This parameter κ controls the additional energy conservation from air cooling/power supply when power management is applied. As the parameter previously considered, there is no additional gain when $\kappa = 0$, with a worsening of the energy savings gains. For higher values of κ, there is a corresponding increase in the energy savings.

Considering only the DPS primitives in the Telecom Italia scenario, the access and core networks show a positive trend based on the value of κ, which is more pronounced than in the metro/transport or particularly in the core



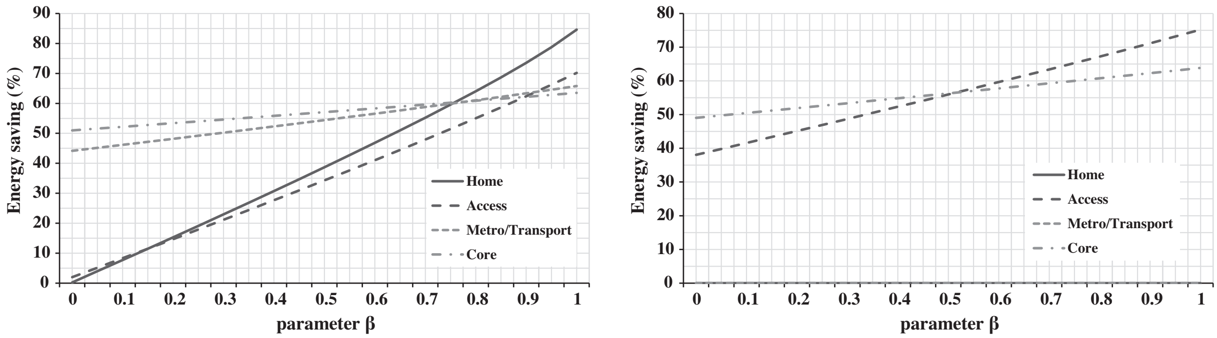
(a) TELIT (b) GRNET

Fig. 23. The estimated energy savings for Telecom Italia (a) and GRNET (b) home, access, metro/transport and core networks during working day profile between the proposed energy model by varying the traffic engineering efficiency (δ parameter) according the BAU scenarios.



(*a*) TELIT (Stand-by primitives) (*b*) GRNET (Stand-by primitives)

Fig. 24. The estimated energy savings for TELIT (a) and GRNET (b) home, access, metro/transport and core networks during working day profile between the proposed energy model by varying the Stand-by efficiency (α parameter) with only Stand-by primitives and the BAU scenarios.



(*a*) TELIT (*b*) GRNET

Fig. 25. The estimated energy savings for Telecom Italia (a) and GRNET (b) home, access, metro/transport and core networks during working day profile between the proposed energy model by varying the performance scaling efficiency (β parameter) and the BAU scenarios.

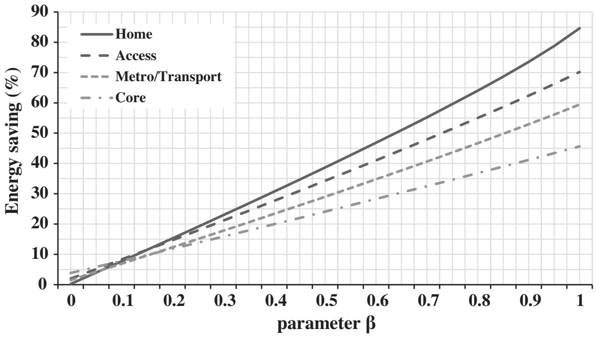
networks (Fig. 22b). Similar considerations are valid for the GRNET scenario.

5.2.8. Evaluation of parameter δ

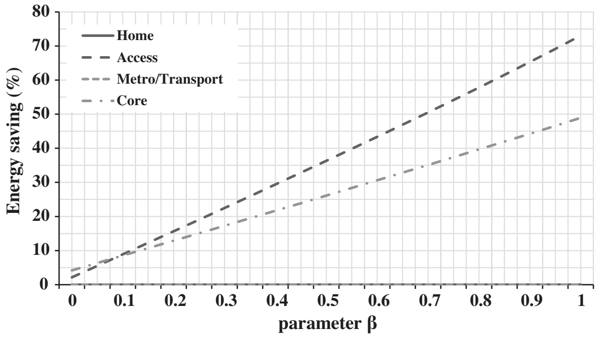
The parameter δ models the efficiency of routing and traffic engineering policies that can be applied to empty network portions during low traffic periods and to allow additional HW to enter standby modes. In this respect, the results in Fig. 22 show how good green traffic engineering and routing policies may increase energy gains by more than 10% (see Fig. 23).

5.2.9. Parameter evaluation conclusions

In this subsection we provide a brief critical analysis of the parameter evaluation results from the previous sections (Sections 5.2.1–5.2.8).



(*a*) TELIT (DPS primitives)



(*b*) GRNET (DPS primitives)

Fig. 26. The estimated energy savings for Telecom Italia (a) and GRNET (b) home, access, metro/transport and core networks during working day profile between the proposed energy model by varying the performance scaling efficiency (β parameter) with only DPS primitives and the BAU scenarios.

The highest energy gains occur when shaping parameters and efficiency degrees approach their ideal values, which may be not so easily achieved with real HW technologies. However, the results clearly identify and characterise the impact of single parameters and degrees, allowing better understanding of which aspects and details of next-generation green devices need to be accurately refined and which could be neglected to effectively reduce the power absorption of tomorrow's network infrastructure.

Among all the analysed parameters, α and β play crucial roles because their variations can affect the overall energy gains by more than 50% and up to 85% depending on the network segments (see Figs. 20 and 25). In more detail, on the one hand, the value of α specifically affects the ISP networks and core and metro/transport segments in the Telco's infrastructure and therefore needs a significant consideration in designing high-speed network nodes. On the other hand, β is clearly the key factor for reducing the enormous amount of energy in home and access networks. Furthermore, the indirect energy savings from the air-cooling and power supply (κ) seem to be remarkable (up to 10–15% of the overall gain – Fig. 21).

A minimal number of AR states (approximately 5) is able to provide energy savings very close to the maximum and, thus, this parameter does not reduce energy consumption of more than approximately 5% in all network segments (Fig. 12). Similar considerations can be deduced for v, where the maximum impact is only approximately 2% (Fig. 14) and only applies for telecom's access and home parts.

Fig. 16 shows that the γ parameter exhibits a remarkable impact (up to 10%) on both telecom's access and home networks. Therefore, LPI power consumption needs to be carefully designed in those devices (e.g., home gateways, DSLAMs) that are usually deployed in such network segments. A similar impact on the same types of network devices is provided by σ (Fig. 18). This double impact of both the σ and γ parameters clearly suggests the importance of LPI primitives in telecom/residential network terminations.

Finally, the design and the efficiency level (δ) of routing and traffic engineering primitives may affect the final gain of the ISP and Telco core, metro and transport segments by more than 15%.

6. Conclusions and future work

The main goal of this work was to quantitatively evaluate the potential energy savings from applying energy efficient techniques, while examining the trade-off between the device's performance and the achieved energy savings.

We introduced a categorisation of the energy aware design space that focused on the existing techniques in the device data plane and we proposed an analytical framework to represent the impact of energy adaptive technologies and solutions for network devices. The proposed energy profile model represents the diverse energy-aware states of the network devices and was evaluated for two example reference scenarios, one of a large-scale Telco (Telecom Italia) and one of a medium size Internet Service Provider (GRNET), to evaluate the impact of each energy-aware technology and the energy saving potentials at the Home, Access, Metro/Transport and Core parts of each network. The considered scenarios included network topologies and device densities, as well as a (daily) characterisation of traffic volumes (measured on the field) that are usually carried by such networks.

The results give an estimate of energy gains that surpass 60% in many cases, while maintaining the same offered quality of service as in the energy-agnostic case. Moreover, the parametric study of the overall gains identified the aspects that need to be considered to maximise the impact when designing green technologies.

Acknowledgements

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